Discrete Mathematics

Lecture notes, Chapters 1–15

Based on Miklós Bóna, *A Walk Through Combinatorics*

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Chapter 1. The Pigeon-Hole Principle

1.1 The basic pigeon-hole principle

Motivation. Comparing the number of objects with the number of places gives existence results; the art is choosing the balls and the boxes.

Theorem 1.1 (Pigeon-hole principle)

Let $n > k$ be positive integers. If n identical balls are placed into k identical boxes, some box receives at least two balls.

Proof. Otherwise every box has 0 or 1 ball, so there are at most $k < n$ balls. ■

Example 1.2

Some element of $7, 77, 777, \ldots$ is divisible by 2023.
Solution. If none of $a_1, \ldots, a_{2023}$ is divisible by 2023, their remainders lie in $\{1, \ldots, 2022\}$, so (Theorem 1.1) $2023 \mid a_j - a_i$ for some $i < j$. But $a_j - a_i = a_{j-i} \cdot 10^i$ and $\gcd(10^i, 2023) = 1$, so $2023 \mid a_{j-i}$, a contradiction.

Example 1.3

In a round-robin chess tournament with n players, at any moment two players have finished the same number of games.
Solution. The possible values are $0, \ldots, n-1$, but 0 and $n-1$ cannot both occur (a player with $n-1$ games has played everybody). So at most $n-1$ values occur; apply Theorem 1.1.

1.2 The generalized pigeon-hole principle

Motivation. Many more balls than boxes force some box to contain many balls.

Theorem 1.4 (Pigeon-hole principle, general version)

Let n, m, r be positive integers with $n > rm$. If n identical balls are distributed into m identical boxes, some box receives at least $r + 1$ balls.

Proof. Otherwise there are at most $rm < n$ balls. ■

Example 1.5

Ten points lie in a unit square. Then two of them are at distance < 0.48, and three of them can be covered by a disk of radius 0.5.
Solution. Nine squares of side $\frac{1}{3}$: one contains two points (Theorem 1.1), at distance $\leq \frac{\sqrt{2}}{3} < 0.48$. Four triangles cut by the diagonals: since $10 > 2 \cdot 4$, one contains three points (Theorem 1.4), and its circumradius is $\frac{1}{2}$.

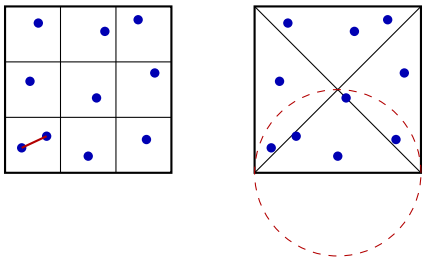


Figure. *Example 1.5: nine squares give two close points; four triangles give three points under a disk of radius* $\frac{1}{2}$.

Example 1.6

Within the last 1000 years, the reader had an ancestor A such that some person P was an ancestor of both parents of A.

Solution. 40 generations of 25 years give a family tree with $2^{41}-1$ nodes. If two nodes are the same person B, then B serves as P (for the nearest common descendant). Otherwise $2^{41}-1$ distinct people lived, but at most $40 \cdot 10^{10} < 2^{41}-1$ did.

Example 1.7 (The handshake party)

Mr. and Mrs. Smith invite four couples. Acquaintances shake hands (nobody with themself or their spouse). Afterwards Mr. Smith observes that, apart from himself, nobody shook hands the same number of times. How many hands did Mrs. Smith shake?

Solution. Vertices = people, edges = handshakes (a **graph**). The nine people other than Mr. Smith have distinct handshake numbers in $\{0,\dots,8\}$; let Y_i have i.

Y_8 missed only the spouse, and Y_0 shook no hands, so Y_8–Y_0 are married.

Y_7 missed two people, Y_0 and Y_1 (whose only handshake was with Y_8), so Y_7–Y_1 are married.

Likewise Y_6–Y_2, Y_5–Y_3, so Mrs. Smith is Y_4: **four** handshakes.

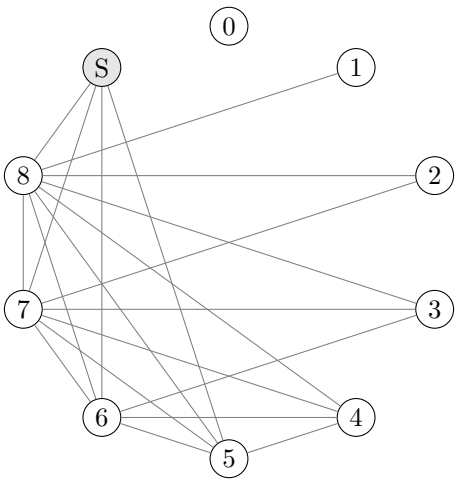


Figure. *The only possible party of Example 1.7 (vertex i shook i hands, S = Mr. Smith); couples* $\{8,0\},\{7,1\},\{6,2\},\{5,3\},\{4,S\}$.

1.3 Selected exercises

Motivation. Book exercises on choosing the boxes.

Exercise 1.11 (Sum or difference divisible by 1000)

Prove that among any 502 positive integers there are two whose sum or difference is divisible by 1000.

Solution.

Step 1 (boxes). With $\bar{r}$ the residue class mod 1000, the 501 boxes

$$\{\bar{0}\}, \quad \{\bar{500}\}, \quad \{\bar{1}, \overline{999}\}, \quad \{\bar{2}, \overline{998}\}, \quad \ldots, \quad \{\overline{499}, \overline{501}\}$$

partition the residue classes.

Step 2 (pigeon-hole). $502 > 501$, so two list members a, b share a box (Theorem 1.1).

Step 3 (conclusion). If $a \equiv b$, then $1000 \mid a - b$; otherwise $a \equiv r$, $b \equiv 1000 - r$, so $1000 \mid a + b$. ■

Exercise 1.15 (Six goals in consecutive games)

A soccer team played 30 games and scored 53 goals, scoring at least one goal in each game.
(a) Prove that there was a sequence of consecutive games in which the team scored exactly six goals.
(b) Prove that the claim of part (a) can fail if the team scored 60 goals, the other conditions unchanged.

Solution.

(a) Let s_k be the goals in games $1, \ldots, k$:

$$1 \le s_1 < s_2 < \cdots < s_{30} = 53.$$

The 60 numbers s_k, $s_k + 6$ lie in $\{1, \ldots, 59\}$, so two are equal (Theorem 1.1); as each list is strictly increasing, $s_j = s_i + 6$ with $j > i$, and games $i + 1, \ldots, j$ have 6 goals.

(b) Goals $1, 1, 1, 1, 1, 7$ repeated five times: 30 games, 60 goals. A block containing a 7 has ≥ 7 goals; otherwise it lies in a run of five 1's and has ≤ 5. ■

Exercise 1.18 (Two 1's in north-east position)

Let $n \ge 2$ and let A be an $n \times n$ matrix with entries 0 and 1 only, at least $2n$ of which are 1. Prove that A contains two entries equal to 1 such that one of them is strictly above and strictly to the right of the other.

Solution. Rows from the top, columns from the left: we need (i', j'), (i, j) with $i' < i$, $j' > j$.

Step 1 (boxes). The $2n - 1$ anti-diagonals $D_s = \{(i, j) : i + j = s\}$, $s = 2, \ldots, 2n$, partition the positions.

Step 2 (pigeon-hole). $2n > 2n - 1$ ones, so two, $(i, j) \ne (i', j')$, lie in one D_s (Theorem 1.1).

Step 3 (conclusion). Then $i \ne i'$, say $i' < i$, and $j' = s - i' > s - i = j$.

Sharpness: the $2n - 1$ ones of the last row and first column contain no such pair (for $(i, 1)$, (n, j) the upper one is to the left). ■

Exercise 1.35 (One number divides another)

(Part (b) of the book's exercise.) Let $n \ge 1$. Prove that among any $n + 1$ distinct integers selected from $[2n]$ there are two such that one is a multiple of the other.

Solution.

Step 1 (boxes). Write uniquely $x = 2^a m$, m odd (the odd part). For $x \in [2n]$, $m \in \{1, 3, \ldots, 2n-1\}$: n values.

Step 2 (pigeon-hole). Two selected integers $x = 2^a m \neq y = 2^b m$ share m (Theorem 1.1).

Step 3 (conclusion). $a \neq b$, say $a < b$; then $y = 2^{b-a} x$.

Sharpness: $n+1, \ldots, 2n$ contain no such pair ($2(n+1) > 2n$). ■

Exercise 1.37 (Multiples of an irrational number)

Let r be an irrational real number. Prove that there is a positive integer n such that the distance of nr from the nearest integer is less than 10^{-10}.

Solution. Write $N = 10^{10}$ and $\{x\} = x - \lfloor x \rfloor \in [0, 1)$ for the fractional part.

Step 1 (boxes). Split $[0, 1)$ into the N half-open intervals $I_a = [\frac{a}{N}, \frac{a+1}{N})$, $a = 0, \ldots, N-1$.

Step 2 (pigeon-hole). Of the $N+1$ numbers $\{kr\}$, $0 \leq k \leq N$, two, $\{kr\}$ and $\{\ell r\}$ with $\ell < k$, share an I_a (Theorem 1.1), so $|\{kr\} - \{\ell r\}| < \frac{1}{N}$.

Step 3 (conclusion). With $n = k - \ell \geq 1$, $M = \lfloor kr \rfloor - \lfloor \ell r \rfloor \in \mathbb{Z}$,

$$|nr - M| = |(kr - \lfloor kr \rfloor) - (\ell r - \lfloor \ell r \rfloor)| = |\{kr\} - \{\ell r\}| < 10^{-10}$$

(and > 0 by irrationality). ■

Chapter 2. The Method of Mathematical Induction

2.1 Weak induction

Motivation. Check the first case and show each case implies the next; the statement itself must be guessed.

The method. To prove $S(m)$ for all natural m:

1. *Initial step:* prove $S(m)$ for the smallest m for which it is defined (usually 0 or 1).
2. *Induction step:* prove that $S(n)$ (the *induction hypothesis*) implies $S(n+1)$.

Validity. If $m+1$ is the smallest failure, it is not the first index, so $S(m)$ holds and the induction step gives $S(m+1)$.

Example 2.1

For all positive integers m,

$$1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}. \tag{2.1}$$

Solution. $m = 1$: both sides are 1. Passing from n to $n+1$ adds $(n+1)^2$ on the left and

$$\frac{(n+1)(n+2)(2n+3) - n(n+1)(2n+1)}{6} = \frac{(n+1)\big[(n+2)(2n+3) - n(2n+1)\big]}{6} = (n+1)^2$$

on the right.

Example 2.2

Let $f(m)$ be the maximum number of regions into which m straight lines divide the plane. Then $f(m) = \frac{m(m+1)}{2} + 1$.
Solution. $f(1) = 2$. The last of $n+1$ lines crosses $k \le n$ others, so it splits $k+1$ regions: $f(n+1) \le f(n) + n + 1$, with equality in general position. Thus $f(n+1) = f(n) + n + 1 = \frac{n(n+1)}{2} + 1 + n + 1 = \frac{(n+1)(n+2)}{2} + 1$.

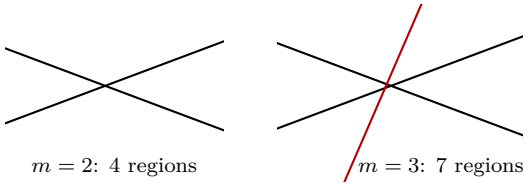


Figure. *Example 2.2: the red line splits 3 old regions, so* $f(3) = f(2) + 3$.

The formula is often guessed from small cases:

Example 2.3

Let $a_0 = 1$ and $a_{n+1} = 3a_n + 1$. Then $a_m = \frac{3^{m+1}-1}{2}$.
Solution. Guessed from $1, 4, 13, 40, 121$. True for $m = 0$, and $a_{n+1} = 3 \cdot \frac{3^{n+1}-1}{2} + 1 = \frac{3^{n+2}-1}{2}$.

A formula like $a_{n+1} = 3a_n + 1$ is a **recurrence relation**; $a_n = \frac{3^{n+1}-1}{2}$ is **explicit** and **closed** (no $\sum$ or $\prod$ over a varying number of terms); $\sum_{i=1}^{n} i$ is explicit but not closed, $\frac{n(n+1)}{2}$ is closed.

Sets. A finite **set** is an unordered collection of distinct objects ($\{1,3,2\} = \{2,1,3\}$); with repetitions, a **multiset**. $B \subseteq A$: every element of B lies in A. The canonical n-element set is $[n] := \{1, 2, \ldots, n\}$.

Theorem 2.4

For every positive integer n, $[n]$ has 2^n subsets.

Proof. $[1]$ has $\emptyset, \{1\}$. Subsets of $[n+1]$ without $n+1$ are the 2^n subsets of $[n]$; those with $n+1$ are $\{n+1\} \cup T$, $T \subseteq [n]$: another 2^n. ■

Pitfalls

"$2n+1$ is even" has a valid induction step but no initial case. "All horses have the same colour" fails because the step from 1 to 2 (two disjoint groups of one horse) is invalid: the step must hold for *every* n.

2.2 Strong induction

Motivation. When a_{n+1} depends on all earlier terms, the induction step may use *all* smaller values.

Example 2.5

Let $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + \cdots + a_n + n + 1$ for $n \geq 0$. Then $a_n = 2^n - 1$.
Solution. True for $n = 0$. If $a_k = 2^k - 1$ for all $k \leq n$, then

$$a_{n+1} = \sum_{k=0}^{n} (2^k - 1) + n + 1 = (2^{n+1} - 1) - (n+1) + n + 1 = 2^{n+1} - 1.$$

The method (strong induction).

1. *Initial step:* prove the statement for the smallest n for which it is defined.
2. *Induction step:* prove that the statement for all integers $< n+1$ implies it for $n+1$.

Validity. As for weak induction, via the smallest failure. From now on $\mathbb{N} = \{0, 1, 2, \ldots\}$.

Example 2.6

Let $f(0) = 1$, $f(1) = 2$ and $f(n+1) = f(n-1) + 2f(n)$ for $n \geq 1$. Then $f(n) \leq 3^n$ for all $n \in \mathbb{N}$.
Solution. True for $n = 0, 1$; then $f(n+1) \leq 3^{n-1} + 2 \cdot 3^n = 7 \cdot 3^{n-1} < 3^{n+1}$ (using both $n-1$ and n).

Variations

One may shift the parameter ($n - 1 \Rightarrow n$) or induct over even or odd integers only.

2.3 Selected exercises

Motivation. Book exercises: the work lies in choosing the statement and what to remove.

Exercise 2.1 (Iterated differences of a polynomial)

For a polynomial p let $\Delta p(n) = p(n) - p(n-1)$, and let $\Delta^k p$ be the result of applying Δ k times ($\Delta^0 p = p$). Prove that if p has degree d, then $\Delta^d p$ is a constant.

Solution. Claim: if $\deg p = d$ with leading coefficient $a \neq 0$, then $\Delta^d p = d!\, a$.

Step 1 (one difference lowers the degree). For $j \geq 1$, by Theorem 4.1

$$n^j - (n-1)^j = n^j - \sum_{i=0}^{j} \binom{j}{i} n^i (-1)^{j-i} = j\, n^{j-1} + (\text{terms of degree} \leq j-2),$$

and for $j = 0$ it is 0. With $p(n) = \sum_{j=0}^{d} c_j n^j$, $c_d = a$,

$$\Delta p(n) = \sum_{j=1}^{d} c_j (n^j - (n-1)^j) = d\, a\, n^{d-1} + (\text{terms of degree} \leq d-2).$$

Step 2 (induction on d). $d = 0$: $\Delta^0 p = a = 0!\, a$. For $d \geq 1$, $q = \Delta p$ has degree $d-1$ and leading coefficient da, so $\Delta^d p = \Delta^{d-1} q = (d-1)! \cdot da = d!\, a$. ■

Exercise 2.2 (A player who lists everybody)

At a tennis tournament every two players played each other exactly once. Afterwards each player listed the names of those he defeated and of those defeated by someone he defeated. Prove that some player listed the names of everybody else.

Solution. *x covers y* if x beat y or x beat some z who beat y. Induction on the number n of players.

Step 1 (initial step). $n = 1$: vacuous.

Step 2 (induction step). Fix X among $n+1$ players; by hypothesis some K covers the other n. If K covers X, done.

Step 3 (the remaining case). Otherwise X beat K and every z beaten by K (else K covers X). For $y \neq X$: if $y = K$ or K beat y, then X beat y; if K beat z who beat y, then X beat z. So X covers everybody. ■

Exercise 2.15 (Arithmetic and geometric means)

Prove by induction that for all nonnegative reals $a_1, \ldots, a_n$,

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Solution. Equivalently $a_1 \cdots a_n \leq A^n$, A the mean; induct on n ($n = 1$: equality).

Step 1 (setting up). For $a_1, \ldots, a_{n+1} \geq 0$ with mean A, reorder so $a_n = \min$, $a_{n+1} = \max$; then $a_n \leq A \leq a_{n+1}$.

Step 2 (replacing two numbers). $b = a_n + a_{n+1} - A \geq a_n \geq 0$ and

$$A\, b - a_n a_{n+1} = (A - a_n)(a_{n+1} - A) \geq 0, \qquad \text{so} \qquad a_n a_{n+1} \leq A\, b.$$

Step 3 (applying the hypothesis). $a_1, \dots, a_{n-1}, b$ have sum nA, so $a_1 \cdots a_{n-1}\, b \le A^n$ and

$$a_1 \cdots a_{n-1} a_n a_{n+1} \le a_1 \cdots a_{n-1} \cdot A\, b \le A^n \cdot A = A^{n+1}.$$

■

Exercise 2.17 (An L-shape of 1's)

(a) Let $m, n \ge 2$ and let A be an $m \times n$ matrix with entries 0 and 1, at least $m + n - 1$ of which are 1. Prove that some entry 1 has another 1 in its row and another 1 in its column.
(b) Does the statement remain true with only $m + n - 2$ entries 1?

Solution.

(a) Otherwise every 1 is alone in its row (R) or, if not, alone in its column (C).

Step 1 (counting). $|R| \le m$, $|C| \le n$, $|R| + |C| \ge m + n - 1$, so $|C| = n$ or $|R| = m$.

Step 2 (case $|C| = n$). Each column's only 1 lies in C, so $R = \emptyset$ and $n \ge m + n - 1$, i.e. $m \le 1$.

Step 3 (case $|R| = m$). Symmetrically $C = \emptyset$ and $n \le 1$.

(b) No: 1's at $(1,1), \dots, (1, n-1)$ and $(2, n), \dots, (m, n)$ ($m + n - 2$ ones); each first-row 1 is alone in its column, each last-column 1 alone in its row. ■

Chapter 3. Elementary Counting Problems

Pigeon-holes and induction draw conclusions *from* numbers; this chapter develops the basic techniques for *finding* them.

3.1 Permutations

Motivation. Order distinct objects; handle repeats by labelling them and dividing out the labellings.

Definition 3.1 (Permutation, factorial)

An arrangement of distinct objects in a linear order, using each exactly once, is a **permutation** of them. The number $n! := n(n-1)\cdots 2\cdot 1$ of permutations of n objects is n **factorial**.

Theorem 3.2

An n-element set has $n!$ permutations.

Proof. n choices for the first position, for each of them $n-1$ for the second, …, 1 for the last. ■

By convention $0! = 1$. **Stirling's formula** (without proof):

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad \text{i.e.} \quad \lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}(n/e)^n} = 1. \tag{3.1}$$

Example 3.3

Flags with three horizontal stripes of the distinct colours red, white, green: $3! = 6$ (RWG, RGW, WRG, WGR, GWR, GRW).

Example 3.4

Five red, three yellow and two white flowers can be planted in a row in $A = \frac{10!}{5!\,3!\,2!} = 2520$ ways.
Solution. Labelling the flowers of each colour, each pattern gives $5!\,3!\,2!$ of the $10!$ labelled rows: $A \cdot 5!\,3!\,2! = 10!$.

A collection with repetitions is a **multiset**; a linear order of all its elements is a **multiset permutation**.

Theorem 3.5 (Multiset permutations)

Let $a_1 + \cdots + a_k = n$ be nonnegative integers. A multiset with a_i objects of type i ($i \in [k]$) has

$$\frac{n!}{a_1!\,a_2!\cdots a_k!}$$

linear orders.

Proof [Supplement]. Label the objects of type i by $1, \ldots, a_i$. Each permutation of the n labelled objects arises from exactly one pair (linear order, labelling), and a fixed order has $a_1!\cdots a_k!$ labellings. So $A \cdot a_1!\cdots a_k! = n!$ (Theorem 3.2). ■

3.2 Strings over a finite alphabet

Motivation. Words count functions, words without repeats count injections, and bijections transfer counts.

Theorem 3.6

There are n^k strings of length k over an n-element alphabet.

Proof. Each of the k positions can independently be filled in n ways (letters may repeat). ■

Example 3.7

There are $10^k - 10^{k-1} = 9 \cdot 10^{k-1}$ positive integers with k digits.

The bijective method. Pair the elements of two sets; if nobody is left over, the sizes agree.

Definition 3.8 (Bijection)

Let X, Y be finite sets and $f : X \to Y$ a function such that (1) $f(a) = f(b) \Rightarrow a = b$, and (2) every $y \in Y$ equals $f(x)$ for some $x \in X$. Then f is a **bijection** from X onto Y; equivalently, each $y \in Y$ has exactly one preimage.

Definition 3.9 (Injection, surjection)

A function satisfying (1) is **one-to-one** (**injective**); one satisfying (2) is **onto** (**surjective**).

Proposition 3.10

If there is a bijection from a finite set X onto a finite set Y, then $|X| = |Y|$.

Proof. The pairs $\{x, f(x)\}$ partition $X \cup Y$ with one element from each set; if there are m pairs, $|X| = |Y| = m$. ■

Example 3.11

An n-element set has 2^n subsets (without induction, cf. Theorem 2.4).
Solution. $B \mapsto$ binary string with i-th digit 1 iff $i \in B$ is a bijection (inverse: positions of the 1's); there are 2^n strings (Theorem 3.6).

Example 3.12

Ten intersections, each with a light or not, lit ones possibly with a gas station: coding A (light, station), B (light only), C (neither) gives 3^{10} possibilities.

Theorem 3.13

For $n \geq k$, the number of strings of length k over an n-element alphabet with no repeated letter is

$$(n)_k := n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Proof. n choices for the first letter, $n-1$ for the second, …, $n-k+1$ for the k-th. ■

Example 3.14

Filling five different cabinet posts from 20 candidates: $(20)_5 = 20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ ways.

3.3 Choice problems

Motivation. Unordered selections are subsets, each counted $k!$ times among ordered ones.

How many tickets guarantee the jackpot of a lottery drawing 5 numbers from [90] (order irrelevant)?

Definition 3.15 (Binomial coefficient)

$\binom{n}{k}$ ("n choose k") denotes the number of k-element subsets of $[n]$; these are the **binomial coefficients** (Chapter 4).

Theorem 3.16

For $0 \le k \le n$: $\displaystyle\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}$.

Proof. Each k-subset appears $k!$ times (Theorem 3.2) among the $(n)_k$ strings of distinct elements (Theorem 3.13): $\binom{n}{k} \cdot k! = (n)_k$. ■

So the lottery needs $\binom{90}{5} = 43\,949\,268$ tickets.

Definition 3.17 (Complement)

For $S \subseteq [n]$, the **complement** S^c is the set of $i \in [n]$ with $i \notin S$.

Proposition 3.18

For $0 \le k \le n$:
(1) $\binom{n}{k} = \binom{n}{n-k}$;
(2) $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. (1) $S \mapsto S^c$ is a bijection from k-subsets to $(n-k)$-subsets (inverse $T \mapsto T^c$); apply Proposition 3.10.

(2) The only 0-subset is $\emptyset$; then use (1). ■

In particular $\binom{0}{0} = 1$, and we set $\binom{n}{k} := 0$ for $k > n$.

Example 3.19

A student must work five days in January, never on two consecutive days. This can be done in $\binom{27}{5}$ ways.
Solution. For $a_1 < \cdots < a_5$ in [31], no two consecutive $\iff$ $1 \le a_1 < a_2 - 1 < a_3 - 2 < a_4 - 3 < a_5 - 4 \le 27$, so $a_i \mapsto a_i - (i-1)$ is a bijection onto the 5-subsets of [27].

Example 3.20

If the five lottery numbers from [90] are drawn *with replacement* and only the multiset counts, $\binom{94}{5}$ tickets are needed.
Solution. $f(b_1,\dots,b_5) = (b_1, b_2+1, b_3+2, b_4+3, b_5+4)$ maps multisets

$$1 \le b_1 \le b_2 \le b_3 \le b_4 \le b_5 \le 90 \tag{3.2}$$

bijectively onto the 5-subsets of [94] ((3.2) iff the shifted values increase strictly in [94]).

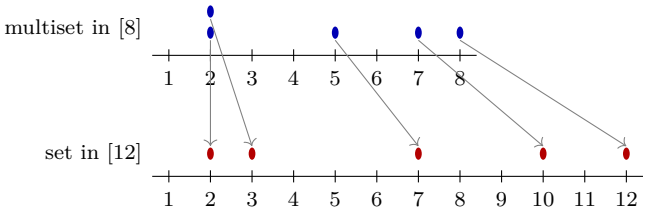


Figure. *Example 3.20 in small: adding* $0,1,2,3,4$ *sends* $\{2,2,5,7,8\}$ *to* $\{2,3,7,10,12\} \subseteq [12]$.

Theorem 3.21

The number of k-element multisets with elements from $[n]$ is $\binom{n+k-1}{k}$.

Proof [Supplement]. For $1 \le b_1 \le \cdots \le b_k \le n$ let $f(b_1,\dots,b_k) := (b_1, b_2+1, \dots, b_k+k-1)$. Since $b_i \le b_{i+1} \iff b_i + (i-1) < b_{i+1} + i$ and $b_k \le n \iff b_k + k - 1 \le n + k - 1$, f is a bijection onto the k-subsets of $[n+k-1]$ (inverse: subtract $0,\dots,k-1$). Apply Proposition 3.10. ■

	Parameters	Formula
Permutations	n distinct objects	$n!$
	a_i objects of type i, $\sum a_i = n$	$\frac{n!}{a_1!\cdots a_k!}$
Lists	k of n distinct objects	$(n)_k = \frac{n!}{(n-k)!}$
	words of length k over n letters	n^k
Subsets	k-element subsets of $[n]$	$\binom{n}{k}$
	k-element multisets from $[n]$	$\binom{n+k-1}{k}$

3.4 Selected exercises

Motivation. Book exercises: read quotients of factorials as counts, and build bijections.

Exercise 3.17 (Products of factorials)

Let $k \ge 1$ and let $b_1,\dots,b_k$ be positive integers whose sum is less than the positive integer n. Prove that $b_1!\,b_2!\cdots b_k! < n!$. Can the statement be made stronger?

Solution.

Step 1 (a count). Let $s = b_1 + \cdots + b_k \le n-1$. By Theorem 3.5, $\frac{s!}{b_1!\cdots b_k!}$ counts a nonempty set of multiset permutations, so it is a positive integer:

$$b_1!\,b_2!\cdots b_k! \le s!.$$

Step 2 (conclusion). $b_1! \cdots b_k! \le s! \le (n-1)! = \frac{n!}{n} < n!$ $(n \ge 2)$.

Stronger: the product is at most $(b_1 + \cdots + b_k)!$, with equality iff $k = 1$ (for $k \ge 2$ two types can be swapped). ■

Exercise 3.26 (Permutations without peaks)

How many permutations of $[n]$ have no entry that is larger than both of its neighbours? (The first and the last entry have only one neighbour and never violate the condition.)

Solution. Let a_n count these *peakless* permutations; we show $a_n = 2^{n-1}$.

Step 1 (where n is). For $n \ge 2$, n exceeds its neighbours, so it is the first or last entry.

Step 2 (removing n). Deleting n leaves a peakless permutation of $[n-1]$ (its neighbour becomes an end entry).

Step 3 (inserting n). Conversely, putting n at the front or end of a peakless q on $[n-1]$ gives two different peakless permutations (the new neighbour of n is smaller than n).

Step 4 (counting). So $a_n = 2a_{n-1}$, $a_1 = 1$, and $a_n = 2^{n-1}$. ■

Exercise 3.57 (Matrices with even lines)

How many $n \times n$ matrices with entries 0 and 1 have an even sum in each row and each column?

Solution. Answer: $2^{(n-1)^2}$. Call such matrices *even*; congruences are mod 2.

Step 1 (the map). $M \mapsto$ its upper left $(n-1)\times(n-1)$ block B; there are $2^{(n-1)^2}$ blocks (Theorem 3.6), so show each B extends uniquely.

Step 2 (forced entries). $m_{in} \equiv \sum_{j<n} b_{ij}$ $(i < n)$ and $m_{nj} \equiv \sum_{i<n} b_{ij}$ $(j < n)$ are forced.

Step 3 (the corner). Row n and column n both require $m_{nn} \equiv \sum_{i,j<n} b_{ij}$, so exactly one m_{nn} works.

Step 4 (conclusion). The map is a bijection (Proposition 3.10). ■

Chapter 4. The Binomial Theorem and Related Identities

Binomial and multinomial coefficients; most proofs count the same objects in two ways (**double counting**).

4.1 The binomial theorem

Motivation. Expanding $(x+y)^n$ chooses x or y from each factor, so its coefficients count subsets.

Theorem 4.1 (Binomial theorem)

For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}. \tag{4.1}$$

Proof. Expanding chooses one summand from each of the n factors; the product is $x^k y^{n-k}$ iff x comes from a k-set of factors, in $\binom{n}{k}$ ways. ■

Theorem 4.2

For every positive integer n: $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$.

Proof. Put $x = -1$, $y = 1$ in (4.1): the left side is $0^n = 0$. ■

Theorem 4.3 (Pascal's rule)

For all nonnegative integers n, k:

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. \tag{4.2}$$

Proof. The right side counts $(k+1)$-subsets S of $[n+1]$. If $n+1 \in S$, the rest of S is a k-subset of $[n]$ (first term); if not, S is a $(k+1)$-subset of $[n]$ (second term). ■

Theorem 4.4

For every nonnegative integer n: $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof. Both sides count subsets of $[n]$ (Theorem 2.4; the right grouped by size). Or put $x = y = 1$ in (4.1). ■

The **Pascal triangle** has $\binom{n}{i}$ as i-th entry of row n (from 0).

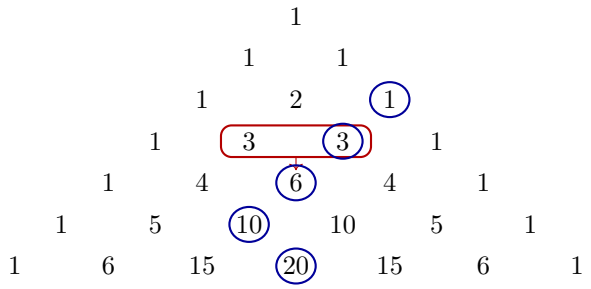


Figure. *Rows 0–6. Red:* $3+3=6$ *(Theorem 4.3). Blue: hockey stick* $1+3+6+10=20$ *(Theorem 4.5).*

Theorem 4.5 (Hockey stick identity)

For all nonnegative integers $k \le n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}. \tag{4.3}$$

Proof. Sort the $(k+1)$-subsets of $[n+1]$ by their largest element $k+i+1$ $(0 \le i \le n-k)$: the rest is one of $\binom{k+i}{k}$ k-subsets of $[k+i]$. ■

Theorem 4.6

For every nonnegative integer n:

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}. \tag{4.4}$$

Proof.

Step 1 (double counting). Count committees with a president: committee first gives $\sum_k k\binom{n}{k}$; president first, then any subset of the other $n-1$, gives $n2^{n-1}$.

Step 2 (calculus, alternative). Differentiate $(x+1)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$: $n(x+1)^{n-1} = \sum_{k\ge1} k\binom{n}{k}x^{k-1}$; put $x=1$. ■

Theorem 4.7 (Vandermonde's identity)

For all positive integers n, m, k:

$$\binom{n+m}{k} = \sum_{i=0}^{k}\binom{n}{i}\binom{m}{k-i}.$$

Proof. Classify k-subsets of $[n+m]$ by the number i of elements in $[n]$: $\binom{n}{i}\binom{m}{k-i}$ each. ■

Theorem 4.8

If $0 \le k \le \frac{n-1}{2}$, then

$$\binom{n}{k} \le \binom{n}{k+1}, \tag{4.5}$$

with equality iff $n = 2k+1$.

Proof. Multiplying $\frac{n!}{k!(n-k)!} \le \frac{n!}{(k+1)!(n-k-1)!}$ by $\frac{k!(n-k-1)!}{n!}$ gives the equivalent $\frac{1}{n-k} \le \frac{1}{k+1}$, i.e. $2k+1 \le n$; equality iff $n-k = k+1$. ■

Corollary 4.9

If $k \geq \frac{n-1}{2}$ (and $k+1 \leq n$), then

$$\binom{n}{k} \geq \binom{n}{k+1}, \tag{4.6}$$

with equality iff $n = 2k+1$.

Proof. With $k' := n-k-1 \leq \frac{n-1}{2}$ and Proposition 3.18(1), the claim is $\binom{n}{k'} \leq \binom{n}{k'+1}$ (Theorem 4.8), with equality iff $n = 2k'+1$, i.e. $n = 2k+1$. ■

$a_0, \ldots, a_n$ is **unimodal** if $a_0 \leq \cdots \leq a_m \geq \cdots \geq a_n$ for some m; so every row of the Pascal triangle is unimodal, peaking in the middle.

4.2 The multinomial theorem

Motivation. Coefficients of $(x_1 + \cdots + x_k)^n$ count multiset permutations.

Example 4.10

$$(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3y^2x + 3y^2z + 3z^2x + 3z^2y + 6xyz. \tag{4.7}$$

Solution. Of the 27 choices, x^2y arises in 3 ways, x^3 in one, xyz in $3 \cdot 2 = 6$.

Definition 4.11 (Multinomial coefficient)

For nonnegative integers with $n = \sum_{i=1}^{k} a_i$,

$$\binom{n}{a_1, a_2, \ldots, a_k} := \frac{n!}{a_1!\, a_2! \cdots a_k!}. \tag{4.8}$$

For $k = 2$, $\binom{n}{a_1, a_2} = \binom{n}{a_1}$.

Theorem 4.12 (Multinomial theorem)

For all nonnegative integers n, k,

$$(x_1 + \cdots + x_k)^n = \sum_{a_1 + \cdots + a_k = n} \binom{n}{a_1, \ldots, a_k} x_1^{a_1} \cdots x_k^{a_k}, \tag{4.9}$$

the sum over all k-tuples of nonnegative integers with sum n.

Proof. $x_1^{a_1} \cdots x_k^{a_k}$ arises from choosing x_i in a_i factors, i.e. from a word of length n with a_i letters x_i (j-th letter = choice in factor j); there are $\binom{n}{a_1, \ldots, a_k}$ (Theorem 3.5). ■

Theorem 4.13

For nonnegative integers with $n = \sum_i a_i$:

$$\binom{n}{a_1, \ldots, a_k} = \binom{n}{a_1}\binom{n-a_1}{a_2}\cdots\binom{n-a_1-\cdots-a_{k-1}}{a_k} \tag{4.10}$$

(the last factor is $\binom{a_k}{a_k} = 1$).

Proof. Choose the positions of x_1 ($\binom{n}{a_1}$ ways), then of x_2 among the remaining $n - a_1$, and so on. ■

4.3 When the exponent is not a positive integer

Motivation. Real exponents give Newton's binomial series, needed for generating functions (Chapter 8).

Definition 4.14 (General binomial coefficients)

For real m and integer $k \geq 0$: $\binom{m}{0} := 1$ and $\binom{m}{k} := \frac{m(m-1)\cdots(m-k+1)}{k!}$ for $k > 0$.

Theorem 4.15 (Binomial theorem, general version)

For every real m and $|x| < 1$,

$$(1+x)^m = \sum_{n\geq 0} \binom{m}{n} x^n.$$

Proof [Supplement]. (The book uses Taylor's theorem.)

Step 1 (convergence). For $m \in \mathbb{N}$ use (4.1). Otherwise $\binom{m}{n} \neq 0$ and $\left|\binom{m}{n+1}x^{n+1} / \binom{m}{n}x^n\right| = \frac{|m-n|}{n+1}|x| \to |x| < 1$, so the series converges on $(-1, 1)$ to f, $f(0) = 1$, differentiable term by term.

Step 2 (differential equation). Using $(n+1)\binom{m}{n+1} = (m-n)\binom{m}{n}$,

$$(1+x)f'(x) = \sum_n \left[(n+1)\binom{m}{n+1} + n\binom{m}{n}\right]x^n = \sum_n m\binom{m}{n}x^n = mf(x).$$

Step 3. $\frac{d}{dx}\left[f(x)(1+x)^{-m}\right] = (1+x)^{-m-1}\left[(1+x)f' - mf\right] = 0$, so $f(x)(1+x)^{-m} \equiv 1$. ■

Example 4.16

$$\frac{1}{\sqrt{1-4x}} = \sum_{n\geq 0}\binom{2n}{n}x^n.$$

Solution. By Theorem 4.15, $(1-4x)^{-1/2} = \sum_n \binom{-1/2}{n}(-4x)^n$ and

$$\binom{-1/2}{n} = \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2n-1}{2})}{n!} = (-1)^n\frac{(2n-1)!!}{2^n n!}, \tag{4.11}$$

with the **semifactorial** $(2n-1)!! = 1\cdot 3\cdots(2n-1)$, $(-1)!! := 1$. So the coefficient is $\frac{2^n(2n-1)!!}{n!} = \frac{(2n-1)!!\,(2\cdot 4\cdots 2n)}{n!\,n!} = \binom{2n}{n}$.

4.4 Selected exercises

Motivation. Book exercises: double counting, consecutive ratios, and reflection of lattice paths.

Exercise 4.4 (Choosing two disjoint subsets)

Let k, m, n be nonnegative integers with $k + m \le n$. Prove that

$$\binom{n}{m}\binom{n-m}{k} = \binom{n}{k}\binom{n-k}{m}.$$

Solution. Count disjoint pairs (M, K) in $[n]$ with $|M| = m$, $|K| = k$.

Step 1 (M first). $\binom{n}{m}\binom{n-m}{k}$.

Step 2 (K first). $\binom{n}{k}\binom{n-k}{m}$.

(Both equal $\binom{n}{m,k,n-m-k}$, Theorem 4.13.) ■

Exercise 4.21 (Log-concavity of binomial coefficients)

(a) Prove that for positive integers $k \le n-1$,

$$\binom{n}{k-1}\binom{n}{k+1} \le \binom{n}{k}^2.$$

(b) (Exercise 4.23.) A sequence $a_0, \dots, a_n$ of positive reals is *log-concave* if $a_{i-1}a_{i+1} \le a_i^2$ for $1 \le i \le n-1$. Prove that a log-concave sequence is unimodal.

Solution.

(a) For $1 \le k \le n-1$,

$$\frac{\binom{n}{k-1}}{\binom{n}{k}} = \frac{k}{n-k+1}, \qquad \frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

so $\binom{n}{k-1}\binom{n}{k+1}/\binom{n}{k}^2 = \frac{k(n-k)}{(k+1)(n-k+1)} < 1$ (strict).

(b)

Step 1 (ratios). $r_i = a_i/a_{i-1}$; dividing $a_{i-1}a_{i+1} \le a_i^2$ by $a_{i-1}a_i > 0$ gives $r_1 \ge r_2 \ge \cdots \ge r_n$.

Step 2 (the peak). Let m be the largest i with $r_i \ge 1$ ($m = 0$ if none). Then $a_{i-1} \le a_i$ for $i \le m$ and $a_{i-1} > a_i$ for $i > m$: $a_0 \le \cdots \le a_m > \cdots > a_n$.

(a) and (b) reprove unimodality of Pascal rows. ■

Exercise 4.24 (Catalan numbers by reflection)

Let C_n be the number of north-eastern lattice paths (steps $(1,0)$ and $(0,1)$) from $(0,0)$ to (n,n) that never go above the diagonal $x = y$. Prove that

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

Solution. There are $\binom{2n}{n}$ paths to (n,n). Let $d = y - x$ (each step changes d by ± 1); a path is *bad* if it visits $d = 1$. Paths from $(-1,1)$ to (n,n) number $\binom{2n}{n-1}$; we biject them with bad paths.

Step 1 (reflection). $\varphi(x,y) = (y-1, x+1)$, the reflection in $y = x+1$, sends $d \mapsto 2-d$, swaps east and north steps, and $\varphi(0,0) = (-1,1)$.

Step 2 (bad paths to paths from $(-1,1)$). Reflect a bad path P up to its first point t with $d = 1$: a path from $(-1,1)$ to (n,n).

Step 3 (the inverse). A path Q from $(-1,1)$ goes from $d = 2$ to $d = 0$, so it visits $d = 1$; reflect it up to the first such point. φ preserves $d \neq 1$, so the first point with $d = 1$ is unchanged and the maps are inverse.

Step 4 (conclusion). $C_n = \binom{2n}{n} - \binom{2n}{n-1} = \binom{2n}{n}\left(1 - \frac{n}{n+1}\right) = \frac{1}{n+1}\binom{2n}{n}$ (ratio from Exercise 4.21(a); $n = 0$ directly). ■

Chapter 5. Partitions

Distributions of n balls into k boxes, each identical or distinct: compositions, set partitions, integer partitions.

5.1 Compositions

Motivation. Identical balls into distinct boxes are ordered sums, counted by balls and walls.

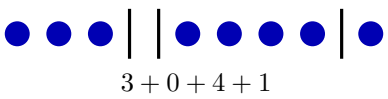
Definition 5.1 (Compositions)

A sequence $(a_1, \ldots, a_k)$ of integers with $a_i \geq 0$ and $a_1 + \cdots + a_k = n$ is a **weak composition** of n into k parts; if all $a_i > 0$, it is a **composition**.

Theorem 5.2

For positive integers n, k, the number of weak compositions of n into k parts is $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$.

Proof. n identical balls into k distinct boxes correspond bijectively to rows of n balls and $k-1$ walls: $\frac{(n+k-1)!}{n!(k-1)!}$ (Theorem 3.5). ■



$3 + 0 + 4 + 1$

Figure. *The weak composition* $(3, 0, 4, 1)$ *as* 8 *balls and* 3 *walls.*

Corollary 5.3

The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Proof [Supplement]. Subtracting 1 from every part is a bijection onto weak compositions of $n-k$ into k parts: $\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$ (Theorem 5.2). ■

Corollary 5.4

A positive integer n has 2^{n-1} compositions (into any number of parts).

Proof.

Step 1 (summing). $\sum_{k=1}^{n} \binom{n-1}{k-1} = 2^{n-1}$ (Corollary 5.3, Theorem 4.4).

Step 2 (doubling, alternative). 1 has one composition. Each composition of n gives two of $n+1$ (add 1 to the first part, or prepend a part 1), and each composition of $n+1$ arises once (first part ≥ 2 or $= 1$). ■

(Weak compositions with unspecified number of parts are infinitely many.)

5.2 Set partitions

Motivation. Distinct balls into identical boxes split $[n]$ into blocks; recurrences follow the largest element.

Definition 5.5 (Set partitions, Stirling numbers of the second kind)

A **partition** of $[n]$ is a collection of nonempty **blocks** such that each element of $[n]$ lies in exactly one block. $S(n,k)$, the **Stirling number of the second kind**, is the number of partitions of $[n]$ into k blocks. $S(n,k) = 0$ for $n < k$; by convention $S(0,0) = 1$ and $S(n,0) = 0$ for $n > 0$.

Example 5.6

$S(n,1) = S(n,n) = 1$ $(n \geq 1)$, $S(n,n-1) = \binom{n}{2}$ $(n \geq 2$; one doubleton).

Example 5.7

$S(4,2) = 7$: $\{1,2,3\}\{4\}$, $\{1,2,4\}\{3\}$, $\{1,3,4\}\{2\}$, $\{2,3,4\}\{1\}$, $\{1,2\}\{3,4\}$, $\{1,3\}\{2,4\}$, $\{1,4\}\{2,3\}$.

No closed formula exists; a single sum follows from the sieve (Chapter 7).

Theorem 5.8

For positive integers $k \leq n$:

$$S(n,k) = S(n-1,k-1) + k\,S(n-1,k). \tag{5.1}$$

Proof. If $\{n\}$ is a block, the rest is one of $S(n-1,k-1)$ partitions; otherwise removing n leaves one of $S(n-1,k)$ partitions, and n joins one of k blocks. ∎

Corollary 5.9

The number of surjections $f : [n] \to [k]$ is $k!\,S(n,k)$.

Proof. Surjections correspond bijectively to partitions into k blocks (the fibres $f^{-1}(i)$) with one of $k!$ labellings by $[k]$. ∎

Corollary 5.10

For every real x and nonnegative integer n:

$$x^n = \sum_{k=0}^{n} S(n,k)(x)_k, \tag{5.2}$$

where $(x)_k = x(x-1)\cdots(x-k+1)$.

Proof. $n = 0$: both sides 1. Both sides are polynomials of degree n, so check $x \in \mathbb{N}$. Then x^n counts functions $[n] \to [x]$ (Theorem 3.6); grouped by image of size k, there are $\binom{x}{k} \cdot k!\,S(n,k) = (x)_k S(n,k)$ (Corollary 5.9). ∎

Definition 5.11 (Bell numbers)

$B(n)$, the n-th **Bell number**, is the number of all partitions of $[n]$; $B(0) = 1$. So $B(n) = \sum_{i=0}^{n} S(n,i)$.

Theorem 5.12

For every nonnegative integer n:

$$B(n+1) = \sum_{i=0}^{n} \binom{n}{i} B(i). \tag{5.3}$$

Proof. If i elements lie outside the block of $n+1$: $\binom{n}{i}$ choices, $B(i)$ partitions of them. Sum over i. ∎

5.3 Integer partitions

Motivation. With nothing distinguishable only box sizes remain; Ferrers diagrams turn identities into pictures.

Definition 5.13 (Integer partitions)

If $a_1 \geq a_2 \geq \cdots \geq a_k \geq 1$ are integers with $a_1 + \cdots + a_k = n$, then $(a_1, \ldots, a_k)$ is a **partition** of the integer n. $p(n)$ denotes the number of partitions of n, and $p_k(n)$ the number with exactly k parts.

Example 5.14

$p(5) = 7$: $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1,1,1,1,1)$.

There is no simple recurrence (but see the Notes). Asymptotically

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{5.4}$$

faster than any polynomial and slower than any c^n, $c > 1$.

The **Ferrers shape** of $(x_1, \ldots, x_k)$ has x_i left-justified boxes in row i. Reflecting in the main diagonal gives the **conjugate**; that of $(4,2,1)$ is $(3,2,1,1)$.

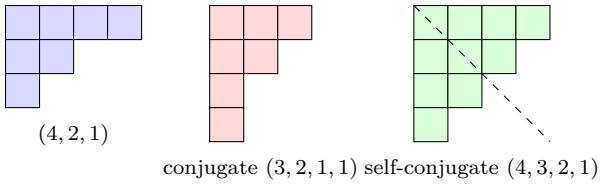


Figure. *Ferrers shapes; conjugation reflects in the diagonal (dashed).*

Definition 5.15 (Self-conjugate)

A partition is **self-conjugate** if it equals its conjugate.

Example 5.16

$(4,3,2,1)$, $(5,1,1,1,1)$ and $(4,2,1,1)$ are self-conjugate.

Theorem 5.17

The number of partitions of n into at most k parts equals the number of partitions of n into parts not larger than k.

Proof. Conjugation swaps shapes with at most k rows and at most k columns. ■

(For $n=5$, $k=2$: $(5),(4,1),(3,2)$ versus $(2,2,1),(2,1,1,1),(1^5)$.)

Theorem 5.18

The number of partitions of n into distinct odd parts equals the number of self-conjugate partitions of n.

Proof.

Step 1 (the map). For self-conjugate π, the first row and column form the **hook** of $2\pi_1-1$ boxes; repeating, the i-th hook has $2\pi_i-(2i-1)$ boxes. $f(\pi)=(2\pi_1-1,2\pi_2-3,\dots)$ has odd, strictly decreasing parts: $(2\pi_i-2i+1)-(2\pi_{i+1}-2i-1)=2(\pi_i-\pi_{i+1})+2\geq 2$.

Step 2 (inverse). For $\alpha=(2a_1-1,\dots,2a_u-(2u-1))$, $f(\pi)=\alpha$ forces a_i boxes in row and column i of the i-th hook, a unique self-conjugate shape with image α. ■

Example 5.19

For self-conjugate $\pi=(6,6,4,3,2,2)$, $f(\pi)=(11,9,3)$.

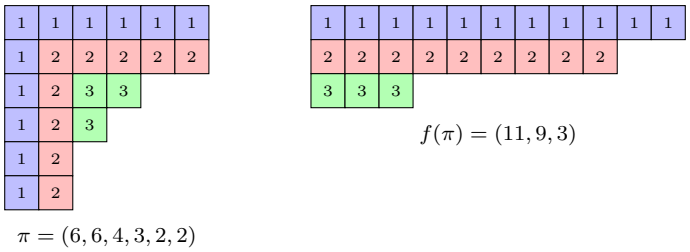


Figure. *Theorem 5.18: hook i becomes row i.*

Theorem 5.20

Let $q(n)$ be the number of partitions of n with all parts at least 2. Then $q(n)=p(n)-p(n-1)$ for $n\geq 2$.

Proof. Appending a part 1 is a bijection from partitions of $n-1$ onto partitions of n with a part 1; the rest are counted by $q(n)$. ■

Type of a set partition. Its block sizes, sorted decreasingly, form a partition of n, the **type**.

Example 5.21

$\{1,5,6\},\{2,7\},\{3,9\},\{4,8\},\{10\}$ has type $(3,2,2,2,1)$.

Theorem 5.22

Let $a = (a_1, \dots, a_k)$ be a partition of n and m_i the multiplicity of i as a part of a. The number of set partitions of $[n]$ of type a is

$$P_a = \frac{\binom{n}{a_1,\dots,a_k}}{\prod_{i\geq 1} m_i!}.$$

Proof. Order a_j balls of colour j ($j \in [k]$) in $\binom{n}{a_1,\dots,a_k}$ ways (Theorem 3.5); same-coloured positions form the blocks, giving every partition of type a. Two colourings give the same partition iff they differ by permuting equal-size colour classes, so each arises $\prod_i m_i!$ times. ■

Example 5.23

There are $\binom{10}{3,2,2,2,1}/(1!\,3!\,1!) = 12600$ set partitions of $[10]$ of type $(3,2,2,2,1)$.

Example 5.24

The partitions of $[2n]$ into blocks of size 2 (perfect matchings) number

$$\frac{\binom{2n}{2,2,\dots,2}}{n!} = \frac{(2n)!}{2\cdot 4\cdots 2n} = \prod_{i=1}^{n}(2i-1) = (2n-1)!!.$$

	Balls	Boxes	Number
Surjections	n distinct	k distinct	$k!\,S(n,k)$
	n distinct	any number, distinct	$\sum_{i=1}^{n} i!\,S(n,i)$
Compositions	n identical	k distinct	$\binom{n-1}{k-1}$
	n identical	any number, distinct	2^{n-1}
Set partitions	n distinct	k identical	$S(n,k)$
	n distinct	any number, identical	$B(n)$
Integer partitions	n identical	k identical	$p_k(n)$
	n identical	any number, identical	$p(n)$
empty boxes allowed (k boxes fixed):			
Functions	n distinct	k distinct	k^n
Weak compositions	n identical	k distinct	$\binom{n+k-1}{k-1}$
Set partitions	n distinct	k identical	$\sum_{i=1}^{k} S(n,i)$
Integer partitions	n identical	k identical	$\sum_{i=1}^{k} p_i(n)$

Notes

With the **pentagonal numbers** $k(3k-1)/2$, $k \in \mathbb{Z}$ (that is $0, 1, 2, 5, 7, 12, \dots$), MacMahon's recurrence reads

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots, \tag{5.5}$$

the i-th sign being $(-1)^{\lfloor i/2 \rfloor}$; e.g. $p(8) = 15 + 11 - 3 - 1 = 22$. Euler: partitions into an even and an odd number of distinct parts are equinumerous unless n is pentagonal (difference one).

5.4 Selected exercises

Motivation. Book exercises on Bell and Stirling numbers.

Exercise 5.2 (Bell numbers versus factorials)

Prove that $B(n) < n!$ for $n \geq 3$.

Solution.

Step 1 (an injection). For a partition π, let σ_π act on each block $\{y_1 < \cdots < y_r\}$ as the cycle $y_1 \mapsto y_2 \mapsto \cdots \mapsto y_r \mapsto y_1$. Its cycles are the blocks, so $\pi \mapsto \sigma_\pi$ is injective: $B(n) \leq n!$.

Step 2 (not surjective). For $n \geq 3$, $\tau = (1\,3\,2)$ (fixing $x \geq 4$) is not some σ_π: π would have block $\{1, 2, 3\}$, but $\sigma_\pi(1) = 2 \neq 3$. So $B(n) \leq n! - 1$. ■

Exercise 5.16 (An alternating sum of Stirling numbers)

Prove that for all positive integers n,

$$\sum_{k=1}^{n} (-1)^k k!\, S(n,k) = (-1)^n.$$

Explain the meaning of this identity in words.

Solution. Put $x = -1$ in Corollary 5.10, with

$$(-1)_k = (-1)(-2)\cdots(-k) = (-1)^k k!$$

and $S(n, 0) = 0$.

Meaning: $k!\, S(n,k)$ counts *ordered* partitions into k blocks (Corollary 5.9); those with an even and an odd number of blocks differ by one, the even ones in excess iff n is even. ■

Exercise 5.34 (Partitions without singleton blocks)

Let $F(n)$ be the number of partitions of $[n]$ with no singleton block. Prove that $B(n) = F(n) + F(n+1)$ for $n \geq 1$, by a bijection.

Solution. Partitions of $[n]$ without singletons ($\mathcal{P}_0$) number $F(n)$; we biject those with a singleton ($\mathcal{P}_1$) onto the set $\mathcal{F}_{n+1}$ of partitions of $[n+1]$ without singletons.

Step 1 (the map). For $\pi \in \mathcal{P}_1$ let $S \neq \emptyset$ be the union of its singletons; $\Phi(\pi)$ replaces them by the block $S \cup \{n+1\}$. So $\Phi(\pi) \in \mathcal{F}_{n+1}$.

Step 2 (the inverse). For $\rho \in \mathcal{F}_{n+1}$ the block of $n+1$ is $T \cup \{n+1\}$, $T \neq \emptyset$; $\Psi(\rho)$ splits T into singletons. $\Psi(\rho) \in \mathcal{P}_1$.

Step 3 (inverse maps). The singletons of $\Psi(\rho)$ are exactly those from T (other blocks have ≥ 2 elements), so $\Phi\Psi = \mathrm{id}$, and $\Psi\Phi = \mathrm{id}$ by construction. Hence $B(n) = F(n) + F(n+1)$. ■

Chapter 6. Cycles in Permutations

A permutation $p = p_1 p_2 \cdots p_n$ of $[n]$ is also the bijection $p : [n] \to [n]$, $p(i) = p_i$.

Example 6.1

312 is the function f with $f(1) = 3$, $f(2) = 1$, $f(3) = 2$.

The **product** fg of permutations of $[n]$ is the composition "first f, then g": $(fg)(i) = g(f(i))$.

Example 6.2

For $f = 312$, $g = 213$: $(fg)(1) = g(3) = 3$, $(fg)(2) = g(1) = 2$, $(fg)(3) = g(2) = 1$, so $fg = 321$.

Example 6.3

$gf = 132 \neq fg$: multiplication is not commutative.

6.1 Cycles in permutations

Motivation. Every permutation splits into cycles; counting by cycles gives Stirling numbers of the first kind.

In $g = 321564$: 2 is fixed, $1 \leftrightarrow 3$ (a 2-cycle), $4 \to 5 \to 6 \to 4$ (a 3-cycle).

Lemma 6.4

For a permutation p of $[n]$ and $x \in [n]$, there is $1 \leq i \leq n$ with $p^i(x) = x$.

Proof. Otherwise $p(x), \ldots, p^n(x)$ lie in $[n] \setminus \{x\}$, so $p^j(x) = p^k(x)$ with $1 \leq j < k \leq n$ (pigeon-hole), and applying p^{-j} gives $x = p^{k-j}(x)$, $1 \leq k - j \leq n - 1$. ■

Definition 6.5 (Cycle)

Let p be a permutation of $[n]$, $x \in [n]$ and i the smallest positive integer with $p^i(x) = x$. Then $x, p(x), \ldots, p^{i-1}(x)$ form an **i-cycle** of p.

Corollary 6.6

Every permutation is the disjoint union of its cycles.

Proof. Every x lies in a cycle (Lemma 6.4). Cycles sharing z both equal the orbit $\{p^m(z) \mid m \in \mathbb{Z}\}$. ■

Example 6.7

321564 has the cycles (31), (2), (564).

In cycle notation $g(i)$ follows i cyclically; e.g. $(241)(35) = (53)(412)$. The **canonical cycle form** starts each cycle with its maximum and orders cycles by increasing first elements: $(412)(53)$. **One-line notation**: $f(1)f(2)\cdots f(n)$.

Example 6.8

(412)(53) in one-line notation is 24513.

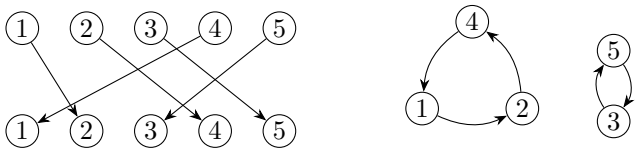


Figure. $24513 = (412)(53)$ *as a function* $[5] \to [5]$ *(left) and as a union of cycles (right).*

Permutations of $[n]$ are called n**-permutations**; their set is the symmetric group S_n.

Theorem 6.9

Let $a_1, \ldots, a_n \geq 0$ with $\sum_i i\, a_i = n$. The number of n-permutations with a_i cycles of length i ($i \in [n]$) is

$$\frac{n!}{a_1!\, a_2! \cdots a_n! \cdot 1^{a_1} 2^{a_2} \cdots n^{a_n}}. \tag{6.1}$$

Proof. Write $[n]$ in a row ($n!$ ways) and bracket a_1 cycles of length 1, then a_2 of length 2, etc. Each permutation of this type arises from $\prod_i i^{a_i} a_i!$ rows (i starting points per i-cycle, $a_i!$ orders of the i-cycles). ■

$(a_1, \ldots, a_n)$ is the (**cycle**) **type** of the permutation.

Example 6.10

There are $(n-1)!$ n-permutations with one cycle (seatings at a round table up to rotation).

Definition 6.11 (Stirling numbers of the first kind)

The number of n-permutations with exactly k cycles is the **signless Stirling number of the first kind** $c(n,k)$; $s(n,k) := (-1)^{n-k} c(n,k)$ is the **Stirling number of the first kind**. $c(n,0) = 0$ for $n > 0$, $c(0,0) = 1$, and $c(n,k) = 0$ for $n < k$.

Theorem 6.12

For positive integers $n \geq k$:

$$c(n,k) = c(n-1, k-1) + (n-1)\, c(n-1,k). \tag{6.2}$$

Proof. Either n is fixed and the rest form $k-1$ cycles ($c(n-1,k-1)$), or the rest form k cycles and n is inserted right after one of the $n-1$ entries ($(n-1)c(n-1,k)$); these choices are distinct and exhaustive. ■

Lemma 6.13

For every positive integer n:

$$\sum_{k=0}^{n} c(n,k) x^k = x(x+1)\cdots(x+n-1). \tag{6.3}$$

Proof. Let $G_n(x) := x(x+1)\cdots(x+n-1) = \sum_k a_{n,k}x^k$ (with $G_0 = 1$). From $G_n = (x+n-1)G_{n-1}$,

$$\sum_k a_{n,k}x^k = \sum_k a_{n-1,k-1}x^k + (n-1)\sum_k a_{n-1,k}x^k.$$

So $a_{n,k}$ satisfies (6.2) with the same initial values, and $a_{n,k} = c(n,k)$ by induction. ■

Replacing x by $-x$ and multiplying by $(-1)^n$:

$$\sum_{k=0}^{n} s(n,k)x^k = (x)_n. \tag{6.4}$$

With (5.2), the lower-triangular matrices $S = (S(n,k))$ and $s = (s(n,k))$ are the change-of-basis matrices between the bases $\{x^k\}$ and $\{(x)_k\}$ of the real polynomials.

Theorem 6.14

The matrices S and s are inverse to each other: $Ss = sS = I$.

Proof. Sums are finite (lower triangular). Substituting (6.4) into (5.2),

$$x^n = \sum_k S(n,k)(x)_k = \sum_j x^j \sum_k S(n,k)s(k,j).$$

By linear independence $\sum_k S(n,k)s(k,j) = \delta_{nj}$: $Ss = I$. Likewise $(x)_n = \sum_j (x)_j \sum_k s(n,k)S(k,j)$ gives $sS = I$. ■

6.2 Permutations with restricted cycle structure

Motivation. Erasing parentheses turns statements about cycles into statements about positions.

Lemma 6.15 (Transition lemma)

For $p \in S_n$ in canonical cycle form, let $g(p)$ be the word obtained by erasing the parentheses, read in one-line notation. Then $g : S_n \to S_n$ is a bijection.

Example 6.16

$g((412)(53)) = 41253$.

Proof of Lemma 6.15. Each $q = q_1 \cdots q_n$ has exactly one bracketing into canonical form: a cycle starts at q_k iff q_k exceeds all earlier leading entries, i.e. (leading entries being cycle maxima) all entries to its left: q_k is a **left-to-right maximum**. These forced brackets give a valid canonical form. ■

Example 6.17

$g^{-1}(4356172) = (43)(5)(61)(72)$: the left-to-right maxima $4, 5, 6, 7$ open the cycles.

t left-to-right maxima of q give t cycles of $g^{-1}(q)$.

Proposition 6.18

For $i \neq j$ in $[n]$, exactly half of all n-permutations contain i and j in the same cycle.

Proof. Relabel so $i = n$, $j = n-1$. In $q = g(p)$, n starts the last cycle, which is n and everything to its right. So they share a cycle iff $n-1$ is right of n in q: half of all q. ■

Lemma 6.19

For $i \in [n]$ and every $k \in [n]$, exactly $(n-1)!$ n-permutations have i in a cycle of length k.

Proof. Let $i = n$. If $q_j = n$ in $q = g(p)$, the cycle of n is $q_j \cdots q_n$, of length k iff $j = n+1-k$: $(n-1)!$ words. ■

So $P(\text{given entry in a } k\text{-cycle}) = \frac{1}{n}$ for every k.

ODD(m), EVEN(m): m-permutations with all cycles of odd, resp. even, length.

Lemma 6.20

$|\text{ODD}(2m)| = |\text{EVEN}(2m)|$ for all positive integers m.

Proof.

Step 1 (the map Φ). $\pi \in \text{ODD}(2m)$ has an even number $2k$ of cycles $C_1, \ldots, C_{2k}$ (canonical order). For $i \leq k$ move the last element of C_{2i-1} to the end of C_{2i} (a singleton disappears): lengths $|C_{2i-1}| - 1$, $|C_{2i}| + 1$, both even. The moved element is at most the leading entry of C_{2i-1}, hence below that of C_{2i}, so canonical form is kept: $\Phi(\pi) \in \text{EVEN}(2m)$.

Step 2 (the inverse). For $\sigma \in \text{EVEN}(2m)$ with cycles $c_1, \ldots, c_h$, work from the right. If the last entry of c_h exceeds the first of c_{h-1}, it was a singleton just before c_h: restore it and continue with c_{h-1}. Otherwise move it back to the end of c_{h-1} and continue with c_{h-2}. A lone remaining cycle's last entry was a singleton. Each step is forced, so the preimage is unique. ■

Example 6.21

$\Phi((4)(513)(726)(8)) = (5134)(72)(86)$.

Example 6.22

$\Phi^{-1}((41)(62)(75)(83)) = (412)(6)(753)(8)$.

Example 6.23

$\Phi^{-1}((21)(53)(64)(87)) = (1)(2)(534)(6)(7)(8)$.

Theorem 6.24

For every positive integer m:

$$|\text{ODD}(2m)| = |\text{EVEN}(2m)| = 1^2 \cdot 3^2 \cdot 5^2 \cdots (2m-1)^2. \tag{6.5}$$

Proof. Count EVEN$(2m)$ (Lemma 6.20). Build p along cycles from 1; when a cycle closes, continue with the smallest unused element.

Step 1. $p(1) \neq 1$: $2m-1$ choices; $p^2(1) \neq p(1)$: $2m-1$ choices (1 closes a 2-cycle).

Step 2. For the $(2i-1)$-st image, $2i-2$ elements are used and the current cycle is even or just starting; the image is neither used nor the start (else an odd cycle closes): $2m-2i+1$ choices. For the $2i$-th image the cycle is odd, so closing is allowed: any of the $2m-(2i-1)$ non-images.

Step 3. $|\mathrm{EVEN}(2m)| = \prod_{i=1}^{m}(2m-2i+1)^2$. ■

For odd n, $\mathrm{EVEN}(n) = \emptyset$, but:

Theorem 6.25

For every positive integer m:

$$|\mathrm{ODD}(2m+1)| = (2m+1)\,|\mathrm{ODD}(2m)| = 1^2 \cdot 3^2 \cdots (2m-1)^2\,(2m+1). \tag{6.6}$$

Proof. By Lemma 6.27 below, Ψ is a bijection $\mathrm{ODD}(2m) \times [2m+1] \to \mathrm{ODD}(2m+1)$; combine with Theorem 6.24. ■

An m-permutation in cycle form has $m+1$ **gap positions**: at the beginning and after each entry.

Example 6.26

$(42)(513)$ has six gap positions: $|(4|2|)(5|1|3|)$.

The map Ψ. For $\pi \in \mathrm{ODD}(2m)$, $k \in [2m+1]$: add 1 to each entry of $\Phi(\pi)$ and insert 1 into the k-th gap (at the front: a new fixed point (1)). One cycle becomes odd and canonical form is kept. Apply Φ^{-1} (relabelled) to the other, even cycles; $\Psi(\pi, k)$ has only odd cycles.

Lemma 6.27

Ψ is a bijection from $\mathrm{ODD}(2m) \times [2m+1]$ onto $\mathrm{ODD}(2m+1)$.

Proof. For $\pi' \in \mathrm{ODD}(2m+1)$: apply Φ to the cycles not containing 1 (an even number of elements); read k as the gap of 1 and delete 1, making its cycle even; subtract 1 and apply Φ^{-1}. Each step inverts one of Ψ. ■

6.3 Selected exercises

Motivation. Book exercises using the cycle decomposition.

Exercise 6.5 (Involutions)

Let $r(n)$ be the number of n-permutations p with $p^2 = \mathrm{id}$, and $r(0) = 1$. Prove that $r(n+1) = r(n) + n\,r(n-1)$ for $n \geq 1$.

Solution. $p^2 = \mathrm{id}$ iff all cycles have length 1 or 2. Split by the cycle of $n+1$.

Step 1 ($n+1$ is fixed). $p|_{[n]}$ is any involution of $[n]$: $r(n)$.

Step 2 ($n+1$ is in a 2-cycle). $(k\ n{+}1)$ for one of n values k, and an involution of $[n]\setminus\{k\}$: $n\,r(n-1)$ ($r(0) = 1$ for $n = 1$). ■

Exercise 6.7 (The average number of cycles)

Prove that n-permutations have on average $H_n = \sum_{i=1}^{n} \frac{1}{i}$ cycles.

Solution.

Step 1 (cycles as a sum over entries). With $\ell_p(i)$ the length of the cycle of i, each cycle contributes $\ell \cdot \frac{1}{\ell} = 1$:

$$\operatorname{cyc}(p) = \sum_{i=1}^{n} \frac{1}{\ell_p(i)}.$$

Step 2 (double counting).

$$\sum_{p \in S_n} \operatorname{cyc}(p) = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k} \left|\{p : \ell_p(i) = k\}\right| = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{(n-1)!}{k} = n!\, H_n,$$

by Lemma 6.19.

Step 3 (conclusion). Divide by $|S_n| = n!$. ■

Exercise 6.21 (Square roots of permutations)

A permutation $p \in S_n$ has a *square root* if $q^2 = p$ for some $q \in S_n$. Prove that p has a square root iff, for every even m, the number of cycles of p of length m is even.

Solution.

Step 1 (square of one cycle). For $c = (x_0\, x_1 \cdots x_{\ell-1})$, $c^2(x_i) = x_{i+2}$ (indices mod ℓ). For odd ℓ, 2 is invertible mod ℓ, so $c^2 = (x_0\, x_2\, x_4 \cdots)$ is one ℓ-cycle; for even ℓ, $c^2 = (x_0\, x_2 \cdots x_{\ell-2})(x_1\, x_3 \cdots x_{\ell-1})$, two $\ell/2$-cycles.

Step 2 (necessity). If $q^2 = p$, p is the union of the squares of the cycles of q (Step 1). An even m-cycle of p comes from a $2m$-cycle of q, which yields two m-cycles; so m-cycles come in pairs.

Step 3 (sufficiency). Pair the cycles of each even length. For an odd cycle c of length $2r+1$, $q = c^{r+1}$ gives $q^2 = c^{2r+2} = c$. For a pair $(y_0 \cdots y_{m-1})$, $(z_0 \cdots z_{m-1})$, $q = (y_0\, z_0\, y_1\, z_1 \cdots y_{m-1}\, z_{m-1})$ gives $q^2(y_i) = y_{i+1}$, $q^2(z_i) = z_{i+1}$.

Step 4 (conclusion). These pieces live on disjoint sets covering $[n]$, so they define $q \in S_n$ with $q^2 = p$. ■

Chapter 7. The Sieve

7.1 Enumerating the elements of intersecting sets

Motivation. Alternately subtracting and adding intersections corrects overcounts exactly, since $\sum_j (-1)^j \binom{s}{j} = 0$.

Example 7.1

14 students play soccer, 17 basketball, and 4 both. Then $14 + 17 - 4 = 27$ play at least one of the two.

Example 7.2

Add 18 hockey players; 4 play soccer and basketball, 3 soccer and hockey, 5 basketball and hockey, and 1 all three. Then $14+17+18-4-3-5+1 = 38$ play at least one (the all-three student is added, subtracted three times each, then added back).

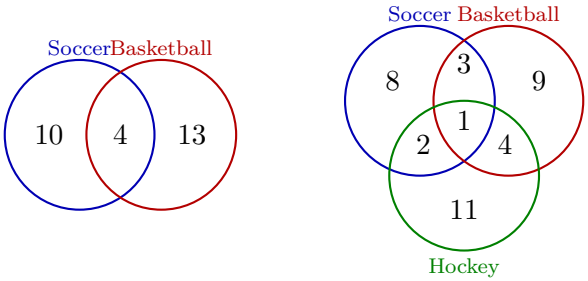


Figure. *Examples 7.1 and 7.2: region counts sum to* 27 *and* 38.

Theorem 7.3 (Sieve formula, inclusion–exclusion)

For finite sets $A_1, \dots, A_n$,

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \sum_{\{i_1,\dots,i_j\}} |A_{i_1} \cap \cdots \cap A_{i_j}|, \tag{7.1}$$

where $\{i_1, \dots, i_j\}$ runs over the j-element subsets of $[n]$.

Proof. For x in the union let $S := \{i \mid x \in A_i\}$, $s := |S| \geq 1$; $x \in A_{i_1} \cap \cdots \cap A_{i_j}$ iff $\{i_1, \dots, i_j\} \subseteq S$, so x is counted

$$s - \binom{s}{2} + \binom{s}{3} - \cdots + (-1)^{s-1} \binom{s}{s} = 1 - (1-1)^s = 1 \tag{7.2}$$

times; elements outside the union are never counted. ∎

7.2 Applications of the sieve formula

Motivation. The sieve counts objects with *none* of several bad properties.

Example 7.4 (Derangements)

How many permutations of $[n]$ have no fixed point (**derangements**; nobody gets their own hat)?

Solution. A_i: permutations fixing i. Each of the $\binom{n}{j}$ intersections of j of them has $(n-j)!$ elements. By (7.1),

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^{n} (-1)^{j-1} \frac{n!}{j!},$$

so

$$D(n) = n! - |A_1 \cup \cdots \cup A_n| = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!}. \tag{7.3}$$

Hence $\frac{D(n)}{n!} \to e^{-1} \approx 37\%$. [Supplement] $D(n)$ is the integer nearest to $n!/e$ ($n \geq 1$): $\left|\frac{n!}{e} - D(n)\right| = n!\left|\sum_{i>n} \frac{(-1)^i}{i!}\right| < \frac{n!}{(n+1)!} \leq \frac{1}{2}$ (alternating, strictly decreasing terms).

Theorem 7.5

For positive integers n, k:

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n.$$

Proof. $k!\,S(n,k)$ counts surjections (Corollary 5.9). Let A_i be the functions missing i; j-fold intersections have $(k-j)^n$ elements, so by the sieve there are $\sum_{j=1}^{k} (-1)^{j-1}\binom{k}{j}(k-j)^n$ non-surjections. Subtract from k^n. ∎

Theorem 7.6 (Möbius inversion on subsets)

Let f, g be real functions on the subsets of $[n]$ with $g(S) = \sum_{T \subseteq S} f(T)$. Then

$$f(S) = \sum_{T \subseteq S} g(T)(-1)^{|S-T|}. \tag{7.4}$$

Proof. The right side is $\sum_{U \subseteq S} f(U) \sum_{U \subseteq T \subseteq S} (-1)^{|S-T|}$. T is determined by $S \setminus T \subseteq S \setminus U$, so the inner sum is $\sum_i (-1)^i \binom{|S-U|}{i} = (1-1)^{|S-U|} = [U = S]$. ∎

7.3 Selected exercises

Motivation. Book exercises on the sieve and Möbius inversion.

Exercise 7.2 (Three teams)

A class of at least two students has three sports teams, and any two students are both members of at least one team. Prove that some team contains at least $\frac{2}{3}$ of the students.

Solution. $n \geq 2$ students, teams T_1, T_2, T_3.

Step 1 (a student in only one team). If a is only in T_1, every b shares a team with a, so $b \in T_1$: T_1 is everybody.

Step 2 (no student in only one team). Every student is in some team ($n \geq 2$), hence in at least two: $|T_1| + |T_2| + |T_3| \geq 2n$. ■

Exercise 7.8 (Euler's function)

Let $\varphi(n)$ be the number of integers in $[n]$ relatively prime to n. Prove that if $p_1, \ldots, p_k$ are the distinct prime divisors of n, then

$$\varphi(n) = n\Big(1 - \frac{1}{p_1}\Big) \cdots \Big(1 - \frac{1}{p_k}\Big).$$

Solution.

Step 1 (the sets). $A_i = \{x \in [n] : p_i \mid x\}$; $\varphi(n) = n - |A_1 \cup \cdots \cup A_k|$.

Step 2 (intersections). With $d_S = \prod_{i \in S} p_i$, $\bigcap_{i \in S} A_i =$ multiples of d_S in $[n]$: n/d_S elements.

Step 3 (sieve). By Theorem 7.3, with $d_\emptyset = 1$,

$$\varphi(n) = n - \sum_{\emptyset \neq S \subseteq [k]} (-1)^{|S|-1} \frac{n}{d_S} = \sum_{S \subseteq [k]} (-1)^{|S|} \frac{n}{d_S} = n \prod_{i=1}^{k} \Big(1 - \frac{1}{p_i}\Big),$$

expanding the product. ■

Exercise 7.12 (The dual Möbius inversion)

Let h, r be real functions on the subsets of $[n]$ with $r(S) = \sum_{T \supseteq S} h(T)$ for all S. Prove that

$$h(S) = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

Solution. Complementation $X \mapsto \bar{X} = [n] \setminus X$ reverses inclusion.

Step 1 (transport). $f(A) = h(\bar{A})$, $g(A) = r(\bar{A})$; with $T = \bar{U}$,

$$g(A) = r(\bar{A}) = \sum_{T \supseteq \bar{A}} h(T) = \sum_{U \subseteq A} h(\bar{U}) = \sum_{U \subseteq A} f(U).$$

Step 2 (apply Theorem 7.6). $f(A) = \sum_{U \subseteq A} g(U)(-1)^{|A-U|}$.

Step 3 (transport back). $A = \bar{S}$, $U = \bar{T}$: $U \subseteq A \iff T \supseteq S$, $A - U = T - S$, so

$$h(S) = f(\bar{S}) = \sum_{T \supseteq S} g(\bar{T})(-1)^{|T-S|} = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

■

Exercise 7.23 (The derangement recurrence)

Give a combinatorial proof of $D(n+1) = n(D(n) + D(n-1))$ for $n \geq 1$, without using (7.3). Here $D(0) = 1$ and $D(1) = 0$.

Solution. For a derangement p of $[n+1]$, $k = p(n+1) \in [n]$: n choices. For fixed k we show $D(n-1) + D(n)$ derangements.

Step 1 ($p(k) = n+1$). $(k\ n{+}1)$ is a 2-cycle and the rest is any derangement of $[n] \setminus \{k\}$: $D(n-1)$.

Step 2 ($p(k) \neq n+1$). Delete $n+1$ from its cycle (length ≥ 3): with $m = p^{-1}(n+1)$, $p'(m) = k$, $p' = p$ elsewhere on $[n]$. p' is a derangement: $m \neq k$ since $p(k) \neq n+1$.

Step 3 (the inverse of Step 2). For a derangement q of $[n]$ and $m = q^{-1}(k)$, set $p(m) = n+1$, $p(n+1) = k$, $p = q$ otherwise: a derangement with $p(k) = q(k) \neq n+1$, and the constructions are inverse. So $D(n)$.

Sum over k. (Check: $D(2) = 1 \cdot (0+1) = 1$.) ■

Exercise 7.37 (No singleton blocks, by the sieve)

Let $F(n)$ be the number of partitions of $[n]$ without singleton blocks. Express $F(n)$ through the Bell numbers.

Solution. Claim: $F(n) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j)$, $B(0) = 1$.

Step 1 (the sets). A_i: partitions with block $\{i\}$; $F(n) = B(n) - |A_1 \cup \cdots \cup A_n|$.

Step 2 (intersections). For $|J| = j$, $\bigcap_{i \in J} A_i$ = singletons of J plus any partition of the other $n-j$: $B(n-j)$ elements, $\binom{n}{j}$ choices of J.

Step 3 (sieve). By Theorem 7.3,

$$F(n) = B(n) - \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} B(n-j) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j).$$

E.g. $F(3) = 5 - 3 \cdot 2 + 3 \cdot 1 - 1 = 1$. Compare Exercise 5.34. ■

Chapter 8. Generating Functions

A generating function stores a whole sequence; operations on it correspond to combinatorial constructions.

8.1 Ordinary generating functions

8.1.1 Recurrence relations and generating functions

Motivation. Summing a recurrence times x^{n+1} packs it into one equation for a power series.

Frogs quadruple yearly and 100 are removed each year: $a_0 = 50$ and

$$a_{n+1} = 4a_n - 100 \qquad (n \geq 0). \tag{8.1}$$

Definition 8.1 (Ordinary generating function)

For a sequence $(f_n)_{n\geq 0}$ of reals, the **formal power series** $F(x) = \sum_{n\geq 0} f_n x^n$ is its **ordinary generating function**.

Formal power series need not converge (e.g. $\sum n!x^n$); equality means equal coefficients.

Multiplying (8.1) by x^{n+1} and summing, with $G(x) = \sum a_n x^n$:

$$G(x) - a_0 = 4xG(x) - \frac{100x}{1-x}, \qquad G(x) = \frac{50}{1-4x} - \frac{100x}{(1-x)(1-4x)}. \tag{8.2}$$

Partial fractions give $\frac{100x}{(1-x)(1-4x)} = \frac{100}{3}\left(\frac{1}{1-4x} - \frac{1}{1-x}\right)$. Since $\frac{1}{1-4x} = \sum 4^n x^n$,

$$a_n = 50 \cdot 4^n - 100 \cdot \frac{4^n - 1}{3}, \tag{8.3}$$

which indeed satisfies $a_0 = 50$ and (8.1).

The method. (1) $G(x) = \sum a_n x^n$. (2) Multiply the recurrence by x^{n+k}, sum over $n \geq 0$. (3) Solve for G. (4) Read off coefficients.

Example 8.2

\$1000 earns 5% interest per year and \$500 is deposited each year: $a_0 = 1000$, $a_{n+1} = 1.05a_n + 500$.

Solution. $G(x) - 1000 = 1.05xG(x) + \frac{500x}{1-x}$, so

$$G(x) = \frac{1000}{1-1.05x} + \frac{500x}{(1-x)(1-1.05x)} = \frac{1000}{1-1.05x} + \frac{10000}{1-1.05x} - \frac{10000}{1-x},$$

and $a_n = 11000 \cdot 1.05^n - 10000$.

Example 8.3

$a_0 = 0$, $a_1 = 1$, $a_{n+2} = 3a_{n+1} - 2a_n$. Multiplying by x^{n+2} and summing: $G - x = 3xG - 2x^2G$. So

$$G(x) = \frac{x}{1-3x+2x^2} = \frac{1}{1-2x} - \frac{1}{1-x}, \qquad a_n = 2^n - 1.$$

8.1.2 Products of generating functions

Motivation. Products convolve coefficients: cut $[n]$ into two intervals with independent structures.

Lemma 8.4

If $A(x) = \sum a_n x^n$, $B(x) = \sum b_n x^n$ and $c_n = \sum_{i=0}^{n} a_i b_{n-i}$, then $A(x)B(x) = C(x) := \sum c_n x^n$.

Proof. $a_i x^i \cdot b_j x^j$ contributes to x^n iff $j = n - i$. ■

Theorem 8.5 (Product formula)

Let a_n (b_n) be the number of structures of a first (second) kind on an n-element set. Let c_n be the number of ways to cut $[n]$ into the intervals $S = \{1, \dots, i\}$ and $T = \{i+1, \dots, n\}$ (possibly empty) and build a structure of the first kind on S and of the second kind on T. Then $A(x)B(x) = C(x)$.

Proof. Cut point i gives $a_i b_{n-i}$ choices; apply Lemma 8.4. ■

By induction, likewise for any fixed number of intervals.

Example 8.6

An n-day semester is split into a first part (k days, one holiday) and a second ($n-k$ days, two holidays). The number of plans is $f_n = \binom{n+1}{4}$.
Solution. $f_n = \sum_k k\binom{n-k}{2}$, with $A(x) = \sum_{k\geq 1} kx^k = \frac{x}{(1-x)^2}$ and $B(x) = \sum_{m\geq 2}\binom{m}{2}x^m = \frac{x^2}{(1-x)^3}$:

$$F(x) = A(x)B(x) = \frac{x^3}{(1-x)^5} = x^3 \sum_{n\geq 0} \binom{n+4}{4} x^n = \sum_{n\geq 3} \binom{n+1}{4} x^n.$$

using $(1-x)^{-5} = \sum_n \binom{n+4}{4}x^n$ (Theorem 4.15 or 3.21).

Example 8.7

With any set of days chosen in each part, each part has $C(x) = \sum 2^k x^k = \frac{1}{1-2x}$, so $F(x) = \frac{1}{(1-2x)^2} = \frac{1}{2}C'(x) = \sum_n (n+1)2^n x^n$, and $g_n = (n+1)2^n$.

Example 8.8

Three parts with any, an odd, and an even number of holidays. An m-set ($m \geq 1$) has 2^{m-1} odd and 2^{m-1} even subsets (Theorem 4.2), giving $\frac{1}{1-2x}$, $\frac{x}{1-2x}$ and $1 + \frac{x}{1-2x} = \frac{1-x}{1-2x}$, and

$$G(x) = \frac{x(1-x)}{(1-2x)^3} = -\frac{1}{4}\cdot\frac{1}{1-2x} + \frac{1}{4}\cdot\frac{1}{(1-2x)^3}.$$

With $(1-2x)^{-3} = \sum_n \binom{n+2}{2}2^n x^n$ this gives $g_n = \frac{1}{4}(\binom{n+2}{2} - 1)2^n = 2^{n-3}n(n+3)$.

Example 8.9

With $p_{\le k}(n)$ the number of partitions of n into parts of size at most k,

$$\sum_{n\ge 0} p_{\le k}(n)x^n = \prod_{i=1}^{k} \frac{1}{1-x^i} = (1+x+x^2+\cdots)(1+x^2+x^4+\cdots)\cdots(1+x^k+x^{2k}+\cdots). \quad (8.4)$$

Solution. Picking x^{ij_i} from factor i with $\sum_i ij_i = n$ corresponds bijectively to the partition with j_i parts i. By Theorem 5.17 this also counts partitions into at most k parts.

Example 8.10

For all partitions,

$$\sum_{n\ge 0} p(n)x^n = \prod_{k\ge 1} \frac{1}{1-x^k}, \quad (8.5)$$

well defined since only the first n factors affect x^n.

Example 8.11 (Euler)

The number $p_{\text{odd}}(n)$ of partitions of n into odd parts equals the number $p_d(n)$ of partitions into distinct parts.
Solution.

$$\sum_n p_{\text{odd}}(n)x^n = \prod_{i \text{ odd}} \frac{1}{1-x^i}, \qquad \sum_n p_d(n)x^n = \prod_{i\ge 1}(1+x^i) = \prod_{i\ge 1}\frac{1-x^{2i}}{1-x^i}.$$

The numerators cancel the even denominators, leaving $\prod_{i \text{ odd}} \frac{1}{1-x^i}$.

8.1.2.1 The Catalan numbers

Let c_n ($c_0 = 1$) count **good sequences** $b_1, \dots, b_{2n} \in \{\pm 1\}$ with $\sum_j b_j = 0$ and all partial sums ≥ 0 (coins into and out of a jar, never negative).

Decomposition. For $n > 0$ let $2i$ be the first return to 0. Days $2i+1, \dots, 2n$ form a good sequence; days $1, \dots, 2i$ form a **very good** one (partial sums positive inside), which minus its first $+1$ and last -1 is a good sequence of length $2(i-1)$. So very good sequences have generating function $xC(x)$, and by the product formula

$$C(x) - 1 = xC(x)\cdot C(x), \qquad \text{i.e.} \quad xC(x)^2 - C(x) + 1 = 0. \quad (8.6)$$

Of the roots $\frac{1\pm\sqrt{1-4x}}{2x}$ only the minus sign is a power series (the plus sign has a pole at 0):

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}. \quad (8.7)$$

By Theorem 4.15, $\sqrt{1-4x} = \sum_n \binom{1/2}{n}(-4x)^n = 1 - 2\sum_{n\ge 1}\frac{1}{n}\binom{2n-2}{n-1}x^n$ (as in Example 4.16), so

$$C(x) = \sum_{n\ge 0}\frac{1}{n+1}\binom{2n}{n}x^n, \qquad c_n = \frac{1}{n+1}\binom{2n}{n}.$$

the **Catalan numbers** $1, 1, 2, 5, 14, 42, \dots$ (Chapter 14).



Figure. *A good sequence as a path above the axis; cutting at the first return gives $C - 1 = xC \cdot C$.*

8.1.3 Compositions of generating functions

Motivation. Cutting into any number of nonempty intervals corresponds to substitution of series.

$F(G(x)) = \sum f_n G^n$ is well defined when G has constant term 0, since then $x^m \mid G^m$.

Definition 8.12 (Composition)

For a formal power series $F(x) = \sum f_n x^n$ and G with constant term 0, $F(G(x)) := \sum_{n \geq 0} f_n G(x)^n$.

Theorem 8.13

Let a_n count structures of some kind on an n-set, with $a_0 = 0$. Let h_n ($h_0 = 1$) count the ways to cut $[n]$ into any number of nonempty intervals and put such a structure on each. Then $H(x) = \frac{1}{1 - A(x)}$.

Proof. $A(x)^k$ counts cuts into k intervals with structures (product formula; nonempty since $a_0 = 0$), and $x^k \mid A^k$, so $H = 1 + \sum_{k \geq 1} A^k = \frac{1}{1-A}$. ∎

(With empty intervals there would be infinitely many cuts.)

Example 8.14

n soldiers in a line are split into nonempty units and a commander is named in each. The number of ways is $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$, with $\alpha, \beta = \frac{3 \pm \sqrt{5}}{2}$.

Solution. $a_k = k$, $A(x) = \frac{x}{(1-x)^2}$, $H(x) = 1 + \frac{x}{1 - 3x + x^2}$; with $\alpha\beta = 1$ the roots of $x^2 - 3x + 1$,

$$\frac{1}{1 - 3x + x^2} = \frac{1}{\sqrt{5}}\left(\frac{\alpha}{1 - \alpha x} - \frac{\beta}{1 - \beta x}\right) = \frac{1}{\sqrt{5}} \sum_n (\alpha^{n+1} - \beta^{n+1}) x^n.$$

So $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$: $1, 3, 8, 21, 55, \ldots$.

Theorem 8.15 (Compositional formula)

Let a_n count structures of a first kind on an n-set ($a_0 = 0$) and b_n structures of a second kind ($b_0 = 1$). Let g_n ($g_0 = 1$) count the ways to cut $[n]$ into nonempty intervals, put a first-kind structure on each interval and a second-kind structure on the set of intervals. Then $G(x) = B(A(x))$.

Proof. k intervals contribute $b_k A(x)^k$ (product formula), so $G = \sum_k b_k A(x)^k = B(A(x))$. ∎

Example 8.16

Split the line into units and choose any subset of units for night duty: $a_k = 1$ $(k \geq 1)$, $b_m = 2^m$, $A = \frac{x}{1-x}$, $B = \frac{1}{1-2x}$,

$$G(x) = \frac{1}{1 - \frac{2x}{1-x}} = \frac{1-x}{1-3x} = 1 + \sum_{n \geq 1} 2 \cdot 3^{n-1} x^n.$$

8.2 Exponential generating functions

8.2.1 Recurrence relations and exponential generating functions

Motivation. Dividing by $n!$ tames fast-growing sequences and absorbs factors $(n+1)$.

Example 8.17

$a_0 = 1$, $a_{n+1} = (n+1)(a_n - n + 1)$ for $n \geq 0$. Then $a_n = n! + n$.

Definition 8.18 (Exponential generating function)

For a sequence $(f_n)_{n \geq 0}$, the formal power series $F(x) = \sum_{n \geq 0} f_n \frac{x^n}{n!}$ is its **exponential generating function** (the constant sequence 1 has e^x).

Solution of Example 8.17. Multiply by $\frac{x^{n+1}}{(n+1)!}$ and sum:

$$\sum_{n \geq 0} a_{n+1} \frac{x^{n+1}}{(n+1)!} = \sum_{n \geq 0} a_n \frac{x^{n+1}}{n!} - \sum_{n \geq 0} (n-1) \frac{x^{n+1}}{n!}, \tag{8.8}$$

i.e. $A(x)(1-x) = 1 + xe^x(1-x)$, so

$$A(x) = \frac{1}{1-x} + xe^x = \sum_{n \geq 0} n! \frac{x^n}{n!} + \sum_{n \geq 1} n \frac{x^n}{n!}.$$

Example 8.19

$f_0 = 0$, $f_{n+1} = 2(n+1)f_n + (n+1)!$. Multiplying by $\frac{x^{n+1}}{(n+1)!}$ and summing: $F(x) = 2xF(x) + \frac{x}{1-x}$. So $F(x) = \frac{x}{(1-x)(1-2x)} = \sum_n (2^n - 1)x^n$, and $f_n = (2^n - 1)\, n!$.

8.2.2 Products of exponential generating functions

Motivation. Exponential products split $[n]$ into two *arbitrary* subsets.

Lemma 8.20

For exponential generating functions $A(x) = \sum a_i \frac{x^i}{i!}$, $B(x) = \sum b_k \frac{x^k}{k!}$ and $c_n = \sum_{i=0}^{n} \binom{n}{i} a_i b_{n-i}$, we have $A(x)B(x) = C(x) := \sum c_n \frac{x^n}{n!}$.

Proof. $a_i \frac{x^i}{i!} \cdot b_j \frac{x^j}{j!} = a_i b_j \binom{i+j}{i} \frac{x^{i+j}}{(i+j)!}$. ∎

Theorem 8.21 (Product formula, exponential version)

Let a_n, b_n count structures of two kinds on n-sets, and let c_n count the ways to split $[n]$ into disjoint subsets $S \cup T = [n]$ (no other restriction) and build a first-kind structure on S and a second-kind structure on T. Then $A(x)B(x) = C(x)$ for the exponential generating functions.

Proof. $|S| = i$: $\binom{n}{i} a_i b_{n-i}$ choices; apply Lemma 8.20. ■

Example 8.22

Split n players into two lined-up groups, the first wearing one of three shirt colours: $a_k = k!\,3^k$, $A = \frac{1}{1-3x}$; $b_m = m!$, $B = \frac{1}{1-x}$;

$$C(x) = \frac{1}{(1-3x)(1-x)} = \frac{3}{2} \cdot \frac{1}{1-3x} - \frac{1}{2} \cdot \frac{1}{1-x} = \sum_n \frac{3^{n+1}-1}{2} x^n,$$

and $c_n = n!\,\frac{3^{n+1}-1}{2}$.

Example 8.23

For fixed k, the exponential generating function of the Stirling numbers of the second kind is

$$S_k(x) := \sum_{n \geq k} S(n,k) \frac{x^n}{n!} = \frac{1}{k!}(e^x - 1)^k. \tag{8.9}$$

Solution. A nonempty block has generating function $e^x - 1$; $(e^x - 1)^k$ counts k *ordered* blocks, each partition $k!$ times. Expanding $(e^x - 1)^k = \sum_i \binom{k}{i} (-1)^i e^{(k-i)x}$ recovers Theorem 7.5.

Differentiation shifts: $\left(\sum a_n \frac{x^n}{n!}\right)' = \sum a_{n+1} \frac{x^n}{n!}$.

Example 8.24

The exponential generating function of the Bell numbers is $B(x) = e^{e^x - 1}$.
Solution. Summing Theorem 5.12 times $\frac{x^n}{n!}$: $B'(x) = B(x)e^x$ (Lemma 8.20), so $\ln B = e^x + C$, and $B(0) = 1$ gives $C = -1$.

8.2.3 Compositions of exponential generating functions

Motivation. Partitioning into structured blocks exponentiates the generating function.

Theorem 8.25 (Exponential formula)

Let a_n count structures on n-sets ($a_0 = 0$) and let h_n ($h_0 = 1$) count the ways to partition $[n]$ into any number of nonempty blocks and put a structure on each block. Then $H(x) = e^{A(x)}$ for the exponential generating functions.

Proof. $A(x)^k$ counts k *ordered* structured blocks (Theorem 8.21), so $A^k/k!$ counts unordered ones; $x^k \mid A^k$, so $H = 1 + \sum_{k \geq 1} \frac{A^k}{k!} = e^{A(x)}$. ■

Integrating $\frac{1}{1-x} = \sum x^n$ termwise gives

$$\sum_{k\geq 1} \frac{x^k}{k} = \ln \frac{1}{1-x}.$$

Example 8.26

Groups at circular tables: $a_k = (k-1)!$ (Example 6.10), $A = \ln \frac{1}{1-x}$, $H = \frac{1}{1-x}$, $h_n = n!$ (a permutation is a set of cycles).

Example 8.27

Partitions into blocks of sizes 3, 4, 9 have exponential generating function $e^{x^3/3!+x^4/4!+x^9/9!}$ (exponential formula with $A = x^3/3!$ etc., combined by the product formula).

Theorem 8.28 (Compositional formula, exponential version)

Let a_n count first-kind structures ($a_0 = 0$) and b_n second-kind structures ($b_0 = 1$). Let g_n count the ways to partition $[n]$ into nonempty blocks, put a first-kind structure on each block and a second-kind structure on the set of blocks. Then $G(x) = B(A(x))$.

Proof. k blocks contribute $b_k \frac{A(x)^k}{k!}$ (Theorem 8.21), so $G = B(A(x))$, defined since $a_0 = 0$. ■

Example 8.29

n cards split into ordered subsets of even size, the subsets in a line: $a_n = n!$ for even $n \geq 2$ (else 0), $A = \frac{x^2}{1-x^2}$; $b_n = n!$, $B = \frac{1}{1-x}$;

$$G(x) = \frac{1}{1-\frac{x^2}{1-x^2}} = \frac{1-x^2}{1-2x^2} = 1 + \sum_{m\geq 0} 2^m x^{2m+2},$$

so $g_n = 2^{n/2-1}\, n!$ for even $n \geq 2$, 0 for odd n.

Chapter 9. The Origins of Graph Theory

Euler (1736): can one walk through Königsberg, returning to the start, crossing each of its seven bridges exactly once?

9.1 The notion of graphs. Eulerian trails

Motivation. Only the pattern of connections matters, and the answer depends only on degree parities.

A **graph** has finitely many **vertices** and **edges**, each joining two vertices; the **degree** of a vertex is the number of edge-ends at it. Multiple edges and **loops** are allowed; a graph with neither is **simple**.

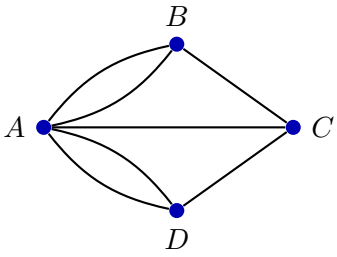


Figure. *Königsberg as a graph: degrees* $5, 3, 3, 3$ *are odd, so no closed Eulerian trail (Theorem 9.2).*

A **trail** is a sequence of distinct edges $e_1 \cdots e_k$ with the end of e_i the start of e_{i+1}; a **walk** may repeat edges. A trail is **closed** if it ends where it starts, **Eulerian** if it uses all edges, and a **path** if it repeats no vertex.

Definition 9.1 (Connected)

G is **connected** if any two vertices are joined by a path.

A **subgraph** H has vertices and edges among those of G; it is **induced** if it contains all edges of G between its vertices. **Connected components** are the maximal connected subgraphs.

Theorem 9.2 (Euler)

A connected graph has a closed Eulerian trail iff all its degrees are even.

Proof.

Step 1 (only if). Along a closed Eulerian trail from S, each visit to $A \neq S$ uses two new edges, so $\deg A = 2a$ (a visits); $\deg S = 2t + 2$ (first and last edge, t intermediate visits).

Step 2 (a first closed trail). Let all degrees be even. From S walk along unused edges. On entering a vertex $\neq S$ an unused edge leaves it (even degree), so we can only stop at S; by finiteness we do, giving a closed trail C_1.

Step 3 (a vertex with unused edges). If $C_1 \neq G$, some vertex V of C_1 has an edge not in C_1: otherwise take $A \notin C_1$; a path from A to C_1 first meets C_1 at some V through an edge not in C_1, a contradiction.

Step 4 (splicing). Removing C_1 keeps degrees even, so by Step 2 there is a closed trail C_2 from V in the remaining edges. C_1 to V, then C_2, then the rest of C_1 is a longer closed trail. Repeat; finiteness gives a closed Eulerian trail. ∎

Corollary 9.3

A connected graph G has an Eulerian trail from S to a different vertex T iff S and T have odd degree and all other degrees are even.

Proof. Add an edge ST to get H. Eulerian S–T trails of G correspond to closed Eulerian trails of H, and H has the degrees of G with $\deg S, \deg T$ raised by 1. Apply Theorem 9.2 to H. ■

Theorem 9.4 (Handshake lemma)

In a graph without loops, the number of vertices of odd degree is even.

Proof. Each edge adds 1 to the degree of each endpoint, so $d_1 + \cdots + d_n = 2e$ is even; hence the number of odd summands is even. ■

9.2 Hamiltonian cycles

Motivation. Visiting every vertex once is NP-complete, but enough edges (Dirac) force it.

A **cycle** is a closed trail repeating no vertex except start = end. A **Hamiltonian cycle** (**path**) is a cycle (path) through all vertices, e.g. a round-table seating where neighbours know each other. Deciding whether one exists is **NP-complete**: a cycle is easy to verify, but no polynomial algorithm is known to certify that none exists (book, Chapter 20).

Theorem 9.5 (Dirac)

Let $n \geq 3$ and let G be a simple graph on n vertices with all degrees at least $n/2$. Then G has a Hamiltonian cycle.

Proof.

Step 1 (connected). If G were disconnected, some component would have $< n/2$ vertices, hence vertices of degree $< n/2$.

Step 2 (a saturated counterexample). Suppose G has no Hamiltonian cycle; add edges while none is created. The result G' has degrees $\geq n/2$, no Hamiltonian cycle, and adding any missing edge creates one; $G' \neq K_n$ since K_n is Hamiltonian.

Step 3 (a Hamiltonian path). For non-adjacent x, y in G', adding xy creates a Hamiltonian cycle through xy, so G' has a Hamiltonian path $x = z_1, \ldots, z_n = y$.

Step 4 (pigeon-hole). Let $A = \{i \in \{2, \ldots, n-1\} \mid xz_i \in E\}$, $B = \{i \in \{2, \ldots, n-1\} \mid z_{i-1}y \in E\}$. As $x \not\sim y$, $|A| = \deg x \geq n/2$, $|B| = \deg y \geq n/2$, so in the $(n-2)$-set $\{2, \ldots, n-1\}$, $|A| + |B| \geq n > n-2$ gives some $i \in A \cap B$.

Step 5. Then $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$ is a Hamiltonian cycle in G', a contradiction. ■

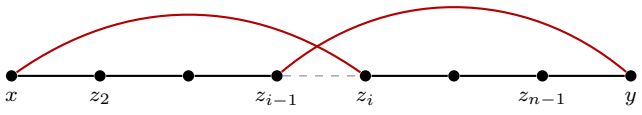


Figure. *Dirac: with $x \sim z_i$ and $z_{i-1} \sim y$ (red), dropping $z_{i-1}z_i$ gives a Hamiltonian cycle.*

9.3 Directed graphs

Motivation. For digraphs Euler's criterion becomes "in-degree = out-degree"; tournaments always have Hamiltonian paths.

In a **digraph** each edge has a direction that trails must follow. G is **strongly connected** if every a has a directed path to every b, and **balanced** if in-degree = out-degree at every vertex. A vertex is **isolated** if both degrees are 0.

Theorem 9.6

A digraph without isolated vertices has a closed Eulerian trail iff it is balanced and strongly connected.

Proof.

Step 1 (necessity). A closed Eulerian trail leaves each vertex as often as it enters it (balanced), and passes through every vertex (strongly connected).

Step 2 (sufficiency) [Supplement]. As in Theorem 9.2: from S walk along unused edges; by balance we can only stop at S, giving a closed trail C_1. If C_1 misses an edge, some V on C_1 has an unused edge: otherwise take $A \notin C_1$ (no isolated vertices) and a directed path from C_1 to A; its first edge leaving C_1 starts at such a V. Removing C_1 keeps balance, so there is a closed trail C_2 from V; splice it into C_1 and repeat. ■

A simple graph is **complete** (K_n) if all $\binom{n}{2}$ pairs are adjacent. An orientation of K_n is a **tournament** (ij an edge if i beat j); there are $2^{\binom{n}{2}}$ on n labelled vertices.

Theorem 9.7

Every tournament has a Hamiltonian path.

Proof. Induction on n; $n \le 2$ is clear. By induction $T - V$ has a Hamiltonian path $h_1 \cdots h_{n-1}$. If $h_i V, V h_{i+1} \in E$ for some i, insert V between them. Otherwise there is k with $V h_j \in E$ for $j \le k$ and $h_j V \in E$ for $j > k$: if $k \ge 1$ put V first, if $k = 0$ put V last. ■

Theorem 9.8 (Camion)

A tournament has a Hamiltonian cycle iff it is strongly connected.

Proof.

Step 1 (only if). A Hamiltonian cycle joins any two vertices by a directed path.

Step 2 (a cycle exists). Let T be strongly connected. With no cycle, $xy, yz \in E$ force $xz \in E$ (else zx closes a 3-cycle), so T is transitive, all edges point right in some ordering, and T is not strongly connected ($n \ge 2$).

Step 3 (a longest cycle). Let $C = y_1 \cdots y_k$ be a longest cycle missing a vertex. Some edge leaves C, say $y_1 x$, $x \notin C$. If $x y_2 \in E$, $y_1 x y_2 \cdots y_k$ is longer; so $y_2 x \in E$, and inductively $y_i x \in E$ for all i.

Step 4. Let $Z = \{z \notin C \mid y_1 z \in E\}$; by Step 3, $y_i z \in E$ for all i, $z \in Z$. Strong connectivity gives $zt \in E$ with $z \in Z$, $t \notin Z$; then $t \notin C$ and $t y_1 \in E$. So $y_k \to z \to t \to y_1 \to \cdots \to y_k$ is a longer cycle, a contradiction. ■

9.4 The notion of isomorphisms

Motivation. Isomorphism makes "same up to relabelling the vertices" precise.

Two graphs on the same labelled vertex set are **identical** if every pair is joined by the same number of edges in both.

Definition 9.9 (Isomorphism)

Graphs G, H are **isomorphic** if there is a bijection f from the vertices of G onto those of H such that X, Y and $f(X), f(Y)$ are joined by the same number of edges, for all X, Y. Such f is an **isomorphism**.

Example 9.10

For the two 4-vertex graphs in the book's Figure 9.8, $f(A) = H$, $f(B) = E$, $f(C) = F$, $f(D) = G$ is an isomorphism; the graphs are not identical (the first contains AB).

Isomorphisms preserve degrees and numbers of cycles and paths of each length, so different degree multisets prove non-isomorphism (equal ones do not prove isomorphism). No efficient general test is known, nor whether the problem is NP-complete; Babai's quasi-polynomial algorithm is a major advance.

9.5 Selected exercises

Motivation. Degree counting, Eulerian trails and longest paths force substructures.

Exercise 9.3 (A 4-cycle from many edges)

Let G be a simple graph on 10 vertices with 28 edges. Prove that G contains a cycle of length 4.

Solution.

Step 1 (cherries). The number of pairs $(v, \{x, y\})$ with $x \neq y$ neighbours of v is $\sum_v \binom{d_v}{2}$.

Step 2 (if there is no 4-cycle). If $v \neq w$ had two common neighbours $x \neq y$, then $v\,x\,w\,y$ would be a 4-cycle. So each $\{x, y\}$ occurs at most once, and $\sum_v \binom{d_v}{2} \leq \binom{10}{2} = 45$.

Step 3 (the degrees force more). $\sum_v d_v = 56$ and by Cauchy–Schwarz $\sum_v d_v^2 \geq \frac{1}{10}\big(\sum_v d_v\big)^2 = 313.6$, so

$$\sum_v \binom{d_v}{2} = \frac{1}{2}\Big(\sum_v d_v^2 - \sum_v d_v\Big) \geq \frac{313.6 - 56}{2} = 128.8 > 45,$$

contradicting Step 2. ∎

Exercise 9.6 (Two orange and two blue edges at each vertex)

Let G be a simple graph in which every vertex has degree 4. Prove that the edges of G can be coloured orange and blue so that every vertex is incident to two orange and two blue edges.

Solution. Colour each component H separately. By Theorem 9.2, H has a closed Eulerian trail $v_0, e_1, v_1, \ldots, e_m, v_m = v_0$.

Step 1 (the colouring). $2m = 4|V(H)|$, so m is even. Colour e_i orange for odd i, blue for even i; put $e_{m+1} = e_1$.

Step 2 (edges at a vertex). Each edge at v has exactly one end at v and occurs once in the trail, so the four edges at v split into the two pairs $\{e_i, e_{i+1}\}$ with $v_i = v$ (for $i = m$ the trail closes via e_1).

Step 3 (conclusion). i and $i+1$ have different parity (also for $\{e_m, e_1\}$, as m is even), so each pair has one edge of each colour. ■

Exercise 9.23 (High minimum degree forces connectivity)

(a) Prove that a simple graph on ten vertices in which every vertex has degree at least five is connected.
(b) Generalize to simple graphs on n vertices.

Solution. (b): a simple graph on n vertices with all degrees $\geq \frac{n-1}{2}$ is connected; $n = 10$ gives (a).

Step 1. For non-adjacent $u \neq v$, $N(u), N(v) \subseteq V \setminus \{u, v\}$ and

$$|N(u)| + |N(v)| \geq n - 1 > n - 2,$$

so some w is adjacent to both, and $u\,w\,v$ is a path.

Step 2. Hence any two vertices are joined by a path of length ≤ 2.

Sharp: for even n, two disjoint copies of $K_{n/2}$ have degrees $\frac{n}{2} - 1$. ■

Exercise 9.46 (A long cycle in a regular graph)

Let $k \geq 2$ and let G be a simple graph in which every vertex has degree k. Prove that G contains a cycle of length at least $k + 1$.

Solution.

Step 1 (a longest path). Let $P = v_0 v_1 \cdots v_m$ be a path with the most vertices.

Step 2 (neighbours of an end). Every neighbour of v_0 lies on P (else P extends). The k distinct neighbours lie among $v_1, \ldots, v_m$, so the largest j with $v_j \sim v_0$ has $j \geq k \geq 2$.

Step 3 (the cycle). $v_0 v_1 \cdots v_j v_0$ is a cycle of length $j + 1 \geq k + 1$.

Only $\min \deg \geq k$ is used; K_{k+1} attains the bound. ■

Chapter 10. Trees

Trees: connected graphs that lose connectivity when any edge is removed.

10.1 Minimally connected graphs

Motivation. Four characterizations describe trees; Joyal's bijection counts labelled ones as n^{n-2}.

Walks contain paths [Supplement]. A shortest walk from x to y is a path: a repeated vertex would let us cut out a portion.

Theorem 10.1

For a connected simple graph G the following are equivalent:
(1) G is **minimally connected**: deleting any edge disconnects it;
(2) G contains no cycle.

Definition 10.2 (Tree)

A connected simple graph satisfying (1), equivalently (2), is a **tree**.

Proof of Theorem 10.1.

Step (1)⇒(2). If a cycle C contains ab, a path using ab reroutes along $C-ab$, so $G-ab$ is connected, contradicting (1).

Step (2)⇒(1). If $G-AB$ is connected, a B–A path P avoids AB, and $AB \cup P$ is a cycle. ■

Corollary 10.3

A connected graph is a tree iff every two vertices are joined by exactly one path.

Proof.

Step (unique paths ⇒ tree). If $G-rs$ were connected, r, s would be joined by rs and by a path avoiding it.

Step (tree ⇒ unique paths) [Supplement]. Let $P \neq Q$ be x–y paths, u the last vertex up to which they agree, and w the first vertex after u on P lying on Q after u (y qualifies). The u–w segments of P and Q meet only at u, w and form a cycle, contradicting Theorem 10.1. ■

Theorem 10.4

Every tree on n vertices has $n-1$ edges. Conversely, every connected graph on n vertices with $n-1$ edges is a tree.

Lemma 10.5

A tree on $n \geq 2$ vertices has at least two vertices of degree 1 (**leaves**).

Proof. Take a longest path p. Another neighbour b of an endpoint a would either extend p ($b \notin p$) or close a cycle with ab ($b \in p$). So both endpoints are leaves. ■

Proof of Theorem 10.4.

Step 1 (trees). Induction on n; $n = 1$ is trivial. A tree T on $n+1$ vertices has a leaf l (Lemma 10.5); $T - l$ is a tree on n vertices with $n - 1$ edges, so T has n.

Step 2 (converse) [Supplement]. Let G be connected, n vertices, $n - 1$ edges. Deleting edges of cycles preserves connectivity and ends with a tree on n vertices, which has $n - 1$ edges by Step 1. So nothing was deleted and G is a tree. ■

A **forest** is a graph whose components are trees.

Proposition 10.6

A forest on n vertices with k components has $n - k$ edges.

Proof. By Theorem 10.4 each component has one more vertex than edges. ■

Counting labelled trees. Trees on $[n]$ number $1, 1, 3, 16$ for $n = 1, \dots, 4$, suggesting n^{n-2} (unlabelled trees: book, Chapter 18).

Theorem 10.7 (Cayley's formula)

There are $A_n = n^{n-2}$ trees with vertex set $[n]$.

Proof (Joyal).

Step 1 (doubly rooted trees). Marking a Start and an End vertex (possibly equal) gives $n^2 A_n$ *doubly rooted trees.* We biject the n^n functions $f : [n] \to [n]$ onto them, so $n^2 A_n = n^n$.

Step 2 (functions to trees). Let $C = \{c_1 < \cdots < c_k\}$ be the elements on cycles of f and $d_i := f(c_i)$. Put $d_1, \dots, d_k$ along a path (Start d_1, End d_k) and add $\{j, f(j)\}$ for $j \notin C$. Iterating f reaches C, so this is connected, with $(k - 1) + (n - k) = n - 1$ edges: a tree (Theorem 10.4).

Step 3 (inverse). For j off the Start–End path, $f(j)$ is the neighbour of j towards the path. On the path read $d_1, \dots, d_k$ from Start to End, sort them as $c_1 < \cdots < c_k$ and set $f(c_i) := d_i$. This recovers f uniquely. ■

Example 10.8

$n = 8$: $f(1) = 3$, $f(2) = 4$, $f(3) = 1$, $f(4) = 5$, $f(5) = 5$, $f(6) = 7$, $f(7) = 8$, $f(8) = 6$. Cycles $(13), (5), (678)$, $C = \{1, 3, 5, 6, 7, 8\}$, $(d_i) = (3, 1, 5, 7, 8, 6)$: the Start–End path is $3 - 1 - 5 - 7 - 8 - 6$, and $2 - 4 - 5$ hangs off it.

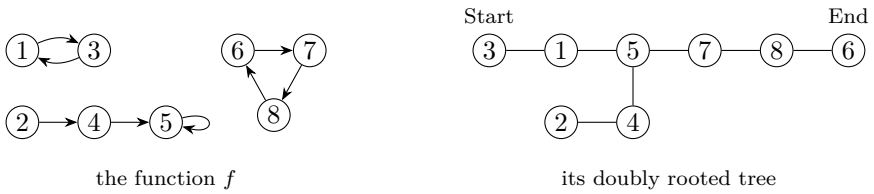


Figure. *Joyal's bijection for Example 10.8: the cyclic part becomes the Start–End path; other elements hang off along $j \mapsto f(j)$.*

A **rooted tree** has one marked root (n^{n-1} on $[n]$); a **rooted forest** is a forest of rooted trees.

Corollary 10.9

There are $(n + 1)^{n-1}$ rooted forests on $[n]$.

Proof. Join every root to a new vertex $n+1$; conversely delete $n+1$ and make its neighbours roots. These are inverse bijections with the $(n+1)^{n-1}$ trees on $[n+1]$ (Theorem 10.7). ■

10.2 Minimum-weight spanning trees. Kruskal's greedy algorithm

Motivation. Greedy is optimal for spanning trees because forests have an exchange property.

A **spanning tree** of connected G is a tree on $V(G)$ using edges of G; one exists (delete cycle edges).

Example 10.10 (Minimum-weight spanning tree)

Given a connected simple graph G and weights $w : E(G) \to \mathbb{R}_+$, find a spanning tree T minimising $w(T) = \sum_{e \in T} w(e)$. (For $n = 20$, $20^{18} > 2.5 \cdot 10^{23}$ trees: over 91 years at 10^9 per second.)

Kruskal's greedy algorithm. Start with $T = \emptyset$; at each step add a minimum-weight edge creating no cycle. T stays a forest; with fewer than $n-1$ edges it is disconnected and some edge of G joins two components, so after $n-1$ steps T is a spanning tree.

Greedy can fail elsewhere: for two disjoint edges of minimum total weight in a path $2, 1, 2$ plus a disjoint edge 10, greedy gives 11, not 4 (Chapter 11).

Lemma 10.11 (Exchange property)

If F, F' are forests on the same vertex set and F has fewer edges than F', then some edge e of F' can be added to F so that $F \cup e$ is still a forest.

Proof. Otherwise every edge of F' lies within a component of F, so F' has at least as many components as F and, by Proposition 10.6, at most as many edges, a contradiction. ■

Theorem 10.12

Kruskal's greedy algorithm finds a minimum-weight spanning tree.

Proof.

Step 1. Let $T = \{t_1, \dots, t_{n-1}\}$ be the greedy tree and $H = \{h_1, \dots, h_{n-1}\}$ a lighter spanning tree, both by increasing weight ($w(t_i)$ increase since eligible edges only shrink).

Step 2. Let i be minimal with $\sum_{j \le i} w(h_j) < \sum_{j \le i} w(t_j)$; minimality gives $w(h_i) < w(t_i)$.

Step 3. By Lemma 10.11 for $T_{i-1} = \{t_1, \dots, t_{i-1}\}$ and $H_i = \{h_1, \dots, h_i\}$, some h_j, $j \le i$, can be added to T_{i-1}; as $w(h_j) \le w(h_i) < w(t_i)$, greedy would not pick t_i, a contradiction. ■

10.3 Adjacency matrices of graphs

Motivation. Matrix powers count walks.

Definition 10.13 (Adjacency matrix)

For a graph G on labelled vertices $1, \dots, n$, the **adjacency matrix** $A = A_G$ has A_{ij} equal to the number of edges between i and j (from i to j for digraphs). It is symmetric for undirected graphs.

Example 10.14

For the undirected graph on A, B, C, D with edges AB, AC, AD, BD:

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Example 10.15

For the digraph with edges $A \to B$, $A \to C$, $B \to D$, $D \to A$:

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Theorem 10.16

$(A^k)_{ij}$ is the number of walks of length k from i to j.

Proof. Induction on k; $k = 1$ is the definition. With $B = A^k$ counting length-k walks, walks of length $k+1$ from i to j with next-to-last vertex z number $b_{iz}a_{zj}$; summing over z gives $(A^{k+1})_{ij}$. ■

Theorem 10.17

A simple graph G on n vertices is connected iff all entries of $(I + A)^{n-1}$ are positive.

Proof. Paths have length $\le n-1$, so G is connected iff for all $i \neq j$ some $(A^k)_{ij} > 0$, $1 \le k \le n-1$. Now $(I+A)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} A^k$ with positive coefficients, nonnegative terms, and positive diagonal from I. ■

10.4 The number of spanning trees of a graph

Motivation. Kirchhoff: spanning trees are counted by a determinant, via Cauchy–Binet.

A **spanning tree** of a digraph is a subgraph whose underlying undirected graph is a spanning tree.

Definition 10.18 (Incidence matrix)

For a loopless digraph with vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$, the **incidence matrix** A is $n \times m$ with $a_{ij} = 1$ if v_i is the start of e_j, -1 if v_i is its end, and 0 otherwise.

Theorem 10.19

Let A be the incidence matrix of a loopless digraph G and A_0 the matrix obtained by deleting any one row. Then the number of spanning trees of G is $\det(A_0 A_0^{\top})$.

Proof. Delete the last row. If $m < n - 1$, G is disconnected and both sides are 0.

Step 1 (claim). For every $(n-1)\times(n-1)$ submatrix B of A_0 (edges forming G'), $|\det B| = 1$ if G' is a spanning tree and $\det B = 0$ otherwise. Induction on n; for $n = 2$, $B = (\pm 1)$.

Step 2 (a leaf other than v_n). If v_i, $i \neq n$, has degree 1 in G', its row has a single entry ± 1; expanding reduces to $G - v_i$, $G' - v_i$, and G' is a spanning tree iff $G' - v_i$ is.

Step 3 (no such leaf). Then G' is not a spanning tree (Lemma 10.5), and some vertex has degree 0 (otherwise $\geq \frac{2(n-1)+1}{2} > n-1$ edges). If it is v_i, $i \neq n$, B has a zero row; if it is v_n, each column has one 1 and one -1, so the rows sum to 0. Either way $\det B = 0$.

Step 4 (Cauchy–Binet). $\det(A_0 A_0^\top) = \sum_B (\det B)^2$ over all $(n-1)\times(n-1)$ submatrices B, which by Step 1 is the number of spanning trees. ■

Theorem 10.20 (Matrix-tree theorem)

Let U be a simple undirected graph with vertices $v_1, \ldots, v_n$ and L_0 the $(n-1)\times(n-1)$ matrix with $\ell_{ii} = \deg v_i$, $\ell_{ij} = -1$ if $v_i v_j$ is an edge, and 0 otherwise $(1 \leq i, j \leq n-1)$. Then U has exactly $\det L_0$ spanning trees.

Proof.

Step 1. Replace each edge by two opposite directed edges (G). Then $(A_0 A_0^\top)_{ii} = 2\deg_U v_i$ and $(A_0 A_0^\top)_{ij} = -2$ for adjacent v_i, v_j, so $A_0 A_0^\top = 2L_0$ and $\det(A_0 A_0^\top) = 2^{n-1}\det L_0$.

Step 2. Each spanning tree of U gives exactly 2^{n-1} spanning trees of G (orient each edge), and all arise so. By Theorem 10.19, $2^{n-1}\tau(U) = 2^{n-1}\det L_0$. ■

Example 10.21

K_n: L_0 has $n-1$ on the diagonal, -1 elsewhere. Adding all rows to the first gives $(1, \ldots, 1)$; adding it to the others gives a triangular matrix with diagonal $1, n, \ldots, n$, so $\det L_0 = n^{n-2}$.

Example 10.22

The **complete bipartite graph** $K_{m,n}$ (parts of sizes m, n, all mn edges between) has $n^{m-1}m^{n-1}$ spanning trees. Delete a vertex of the n-part: L_0 has diagonal blocks nI_m, mI_{n-1} and -1 between parts. Adding all rows to the first gives $(1, \ldots, 1, 0, \ldots, 0)$; adding it to the last $n-1$ rows makes it block triangular with determinant $n^{m-1}m^{n-1}$.

Theorem 10.23 (Matrix-tree theorem, eigenvalue version)

Let L be the $n \times n$ **Laplacian** of U (defined like L_0 but over all n vertices), with eigenvalues $\lambda_1, \ldots, \lambda_{n-1}, \lambda_n = 0$. Then the number of spanning trees of U is $\frac{1}{n}\lambda_1\lambda_2\cdots\lambda_{n-1}$.

Proof [Supplement]. Rows of L sum to 0, so $\lambda_n = 0$ and $\det(xI - L) = x\prod_{i<n}(x - \lambda_i)$ has x-coefficient $(-1)^{n-1}\lambda_1\cdots\lambda_{n-1}$. Expanding, this coefficient is also $(-1)^{n-1}$ times the sum of the n principal $(n-1)\times(n-1)$ minors, each equal to $\tau(U)$ by Theorem 10.20 (relabel with v_k last). So $\lambda_1\cdots\lambda_{n-1} = n\,\tau(U)$. ■

If U is d-**regular**, $L = dI - A$, so L has eigenvalues $d - \alpha_i$.

Example 10.24

K_n: $A + I = J$ has eigenvalues $n, 0, \ldots, 0$, so A has $n-1, -1, \ldots, -1$ and $L = (n-1)I - A$ has $0, n, \ldots, n$; Theorem 10.23 gives $\frac{1}{n}n^{n-1} = n^{n-2}$.

10.5 Selected exercises

Motivation. Tree characterizations at work: degree sequences, complements, and the cut property.

Exercise 10.1 (Degree sequences of trees)

Let $n \geq 2$ and let $a_1 \geq a_2 \geq \cdots \geq a_n$ be positive integers with $a_1 + \cdots + a_n = 2n - 2$. Prove that there is a tree on n vertices whose vertices have degrees $a_1, \ldots, a_n$.

Solution. Induction on n, for any order of the a_i.

Step 1 (initial step). $n = 2$: $a_1 = a_2 = 1$, a single edge.

Step 2 (a 1 and a part ≥ 2). Let $n \geq 3$. All $a_i \geq 2$ would give sum $\geq 2n$, so say $a_n = 1$; all $a_i = 1$ would give $n < 2n - 2$, so say $a_1 \geq 2$.

Step 3 (reduction). $a_1 - 1, a_2, \ldots, a_{n-1}$ are positive with sum $2(n-1) - 2$, so by induction a tree T' on $v_1, \ldots, v_{n-1}$ has these degrees.

Step 4 (adding a leaf). Add v_n and $v_1 v_n$: connected, n vertices, $n-1$ edges, so a tree (Theorem 10.4) with degrees $a_1, \ldots, a_n$. ■

Exercise 10.10 (A graph or its complement is connected)

Prove that for every simple graph G on $[n]$, at least one of G and its complement $\bar{G}$ is connected. Give an example where both are connected.

Solution. Let G be disconnected and $x \neq y$.

Step 1 (different components). Then $xy \in E(\bar{G})$.

Step 2 (same component). Take z in another component; $x\,z\,y$ is a path in $\bar{G}$.

So $\bar{G}$ is connected. Example: the path $1\,2\,3\,4$ has complement $13, 14, 24$, the path $3\,1\,4\,2$. ■

Exercise 10.24 (The cut property)

A *cut* in a connected graph G is a minimal set S of edges whose removal splits G into two components. Let w be a weight function on the edges, let S be a cut, and let $e \in S$ satisfy $w(e) < w(f)$ for all other $f \in S$. Prove that every minimum-weight spanning tree of G contains e.

Solution.

Step 1 (the shape of a cut). Let V_1, V_2 be the vertex sets of the components of $G - S$. Edges between V_1, V_2 lie in S. Conversely, if $f \in S$ had both ends in one V_i, then $S \setminus \{f\}$ would still split G into two components, contradicting minimality. So S is exactly the set of V_1–V_2 edges.

Step 2 (exchange). Let T be a minimum-weight spanning tree with $e = uv \notin T$, $u \in V_1$, $v \in V_2$. The unique u–v path in T (Corollary 10.3) has an edge f from V_1 to V_2; $f \in S$, $f \neq e$, so $w(f) > w(e)$.

Step 3 (a cheaper tree). $T - f$ has $n-2$ edges, hence two components (Proposition 10.6), separating u, v (else a second u–v path). So $T' = T - f + e$ is connected with $n - 1$ edges, a spanning tree (Theorem 10.4), and

$$w(T') = w(T) - w(f) + w(e) < w(T),$$

a contradiction. Hence $e \in T$. ■

Chapter 11. Coloring and Matching

11.1 Introduction

Motivation. Assigning frequencies so nearby towers differ is a vertex colouring.

Four frequencies, ten towers; towers within 50 miles must differ. Join such towers by an edge and colour vertices by frequency.

Proposition 11.1

The four frequencies can be assigned to the ten towers iff the vertices of the graph can be coloured with four colours so that no two adjacent vertices have the same colour.

Proof. An assignment is a 4-colouring, and "towers within 50 miles differ" is "adjacent vertices differ". ■

Definition 11.2 (Chromatic number)

The **chromatic number** $\chi(H)$ is the least k such that the vertices of H can be coloured with k colours with adjacent vertices coloured differently (a **proper colouring**). H is **k-colourable** if $\chi(H) \leq k$.

Example 11.3

$\chi(C_5) = 3$: two colours cannot alternate around an odd cycle; red, blue, red, blue, green works.

We may assume graphs connected and simple (components colour independently; multiple edges add no constraints).

11.2 Bipartite graphs

Motivation. Two colours suffice iff there is no odd cycle; past $n^2/4$ edges, triangles appear in bulk.

Definition 11.4 (Bipartite)

A 2-colourable graph is **bipartite**: its vertex set splits into **colour classes** A, B such that every edge has one end in A and one in B.

Trees (colour by distance parity) and even cycles are bipartite; odd cycles are not.

Theorem 11.5 (König)

A graph is bipartite iff it contains no cycle of odd length.

Proof.

Step 1 (only if). Colours alternate along $A_1 \cdots A_{2m+1}$, so the adjacent A_1, A_{2m+1} agree.

Step 2 (colouring by distance). Let G (wlog connected) have no odd cycle. Fix V; colour W red if $d(V, W)$ is even, blue if odd. An edge PQ inside one colour plus shortest paths $V \to P$, $V \to Q$ gives a closed walk of odd length.

Step 3 (odd closed walks contain odd cycles) [Supplement]. Induction on length: without a repeated vertex it is a cycle; otherwise split at a repeated vertex into two shorter closed walks with odd total length, one of them odd. This contradicts the assumption. ■

Theorem 11.6

A simple bipartite graph on n vertices has at most $\frac{n^2}{4}$ edges if n is even and at most $\frac{n^2-1}{4}$ if n is odd.

Proof. An extremal G with classes of sizes $a, n-a$ has all $a(n-a)$ cross edges, maximised at $a = \lfloor n/2 \rfloor$. ■

The extremal graphs are the **complete bipartite graphs** $K_{a,b}$.

Lemma 11.7 (Mantel)

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains a triangle.

Proof.

Step 1 ($m = 2$). $H \subseteq K_4$ with ≥ 5 edges is not bipartite (Theorem 11.6), so it has an odd cycle, on 4 vertices a triangle.

Step 2 (induction). Take an edge FG. If $\deg F + \deg G > 2m$, then F, G have at least $2m-1$ neighbours among the other $2m-2$ vertices (besides each other), hence a common neighbour T: triangle FGT. Otherwise deleting F, G removes at most $2m-1$ edges, leaving $2m-2$ vertices and $\geq (m-1)^2+1$ edges; apply induction. ■

Theorem 11.8

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains at least m triangles.

Proof. Assume exactly m^2+1 edges; induct on m ($m = 2$: K_4 minus an edge has 2 triangles). Lemma 11.7 gives a triangle ABC. Let $2m-3+x$ edges join ABC to the other $2m-3$ vertices; by pigeon-hole at least x outside vertices send two edges to ABC, giving $\geq x$ triangles. Let R be the graph on the outside vertices.

Step 1 ($x \geq m-1$). Done.

Step 2 ($1 \leq x < m-1$). At most $3m-5$ edges meet ABC from outside, so R has $\geq m^2+1-(3m-5)-3 = (m-1)(m-2)+1$ edges on $2m-3$ vertices. Deleting a minimum-degree vertex leaves R' on $2m-4$ vertices with more than $(m-1)(m-2)\frac{2m-4}{2m-3} = (m-2)^2\frac{2m-2}{2m-3} > (m-2)^2$ edges. Induction gives $m-2$ triangles, plus $x \geq 1$.

Step 3 ($x \leq 0$). At most $2m-3$ edges join ABC to R, so R has $\geq (m-1)^2$ edges. With no such edge, R has $m^2-2 \geq (m-1)^2+1$ edges and R plus any vertex of ABC has $2m-2$ vertices. Otherwise add to R a vertex of ABC adjacent to R: $2m-2$ vertices, $\geq (m-1)^2+1$ edges. Induction gives $m-1$ further triangles. ■

11.3 Matchings in bipartite graphs

Motivation. Hall's obviously necessary condition is also sufficient; augmenting paths certify maximality.

Jobs and applicants form the classes, qualifications the edges.

Proposition 11.9

All m openings can be filled simultaneously iff the job–applicant graph has m vertex-disjoint edges.

Proof. A complete hiring is exactly a set of m pairwise vertex-disjoint edges "job–person hired". ■

Definition 11.10 (Matching)

A set S of edges of G, no two sharing a vertex, is a **matching** (independent set of edges). It is **perfect** if it covers every vertex.

Definition 11.11 (Matching X into Y)

For a bipartite $G = (X, Y)$, a matching covering all of X is a **perfect matching of X into Y**; we say X **can be matched into Y**.

11.3.1 Bipartite graphs with perfect matchings

For $T \subseteq X$, $N(T) = N_G(T)$ is the set of neighbours of T.

Proposition 11.12

If X can be matched into Y, then $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof. The matching maps T injectively into $N(T)$. ■

Theorem 11.13 (Hall's theorem)

X can be matched into Y iff $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof (Halmos–Vaughan). "Only if" is Proposition 11.12. "If": induct on $|X|$; $|X| \le 1$ is trivial.

Step 1 (strict surplus). If $|T| < |N(T)|$ for all nonempty proper $T \subset X$, take an edge xy and $G' = G - x - y$: $|N_{G'}(A)| \ge |N_G(A)| - 1 \ge |A|$ for $\emptyset \ne A \subseteq X - x$. Match $X - x$ into $Y - y$ by induction; add xy.

Step 2 (a tight set). Otherwise $|B| = |N(B)|$ for some nonempty proper $B \subset X$. Let $G_1 = G[B \cup N(B)]$ and $G_2 = G - (B \cup N(B))$. Hall's condition holds in G_1 ($N_{G_1}(T) = N_G(T)$), and in G_2, since $N_G(U \cup B) = N_{G_2}(U) \sqcup N_G(B)$ for $U \subseteq X - B$,

$$|N_{G_2}(U)| = |N_G(U \cup B)| - |N_G(B)| \ge |U| + |B| - |B| = |U|.$$

Induction matches B onto $N(B)$ and $X - B$ into $Y - N(B)$. ■

A matching is **maximal** if no edge can be added, **maximum** if none is larger. Maximum implies maximal, not conversely (the middle edge of a 3-edge path).

Definition 11.14 (Alternating path)

For a matching M, a path $v_1 v_2 \cdots v_r$ is **M-alternating** if its edges alternate between M and not M.

Definition 11.15 (Augmenting path)

An M-alternating path is **M-augmenting** if it starts and ends at vertices not covered by M.

$M \mapsto M \triangle P$ for an augmenting path P adds one edge.



Figure. *An M-augmenting path (bold: M); swapping bold and dashed edges enlarges the matching.*

Theorem 11.16 (Berge)

A matching M in a simple graph is maximum iff there is no M-augmenting path.

Proof. "Only if": an augmenting path enlarges M. "If": for a maximum M', the components of $M \triangle M'$ are M/M'-alternating paths and even cycles. A path with more M'-edges would be M-augmenting, one with more M-edges M'-augmenting. So each component is balanced and $|M| = |M'|$. ■

11.3.2 Stable matchings in bipartite graphs

n candidates rank n jobs, and each hiring manager ranks all candidates.

Definition 11.17 (Stable matching)

In a complete bipartite $G(X, Y)$ where every vertex ranks the other side, a perfect matching M is **stable** if there are no $x \in X$, $y' \in Y$ such that (a) $xy \in M$ but x prefers y' to y, and (b) $x'y' \in M$ but y' prefers x to x' (no pair would both rather be together).

Gale–Shapley (deferred acceptance). Each unmatched candidate applies to the best job not yet tried; each manager tentatively keeps the best applicant so far and rejects the rest. It ends with everyone placed (an unplaced candidate always has an untried job).

The result is stable. Let $xy, x'y' \in M$, x prefer y', y' prefer x to x'. Then x applied to y' earlier, and y' either accepted and later dropped x or rejected x outright, each time for someone better; its final x' is at least as good, a contradiction. Unequal sides: add fictitious members ranked last. In non-bipartite graphs stable matchings need not exist.

Remark

Remark[Algorithmic note] $O(n^2)$ applications: each candidate applies to each job at most once.

11.4 More than two colours

Motivation. Turán: more edges than the balanced complete k-partite graph force K_{k+1}.

Write $n = tk + r$, $0 \le r \le k-1$; split n vertices into r classes of size $t+1$ and $k-r$ of size t, joining vertices in different classes. This **complete k-partite (Turán) graph** H is k-colourable with

$$T(n,k) = \frac{k-1}{2k}\,n^2 - \frac{r(k-r)}{2k} \tag{11.1}$$

edges. [Supplement] *Reason:* $\frac{1}{2}(n^2 - r(t+1)^2 - (k-r)t^2)$ with $n = tk + r$ gives (11.1).

Theorem 11.18 (Turán)

A simple graph on n vertices with more than $T(n,k)$ edges contains K_{k+1}; in particular it is not k-colourable.

Proof.

Step 1. Let G be K_{k+1}-free on n vertices with the most edges; show $e(G) \le T(n,k)$ by induction on $t = \lfloor n/k \rfloor$. For $t = 0$, $T(n,k) = \binom{n}{2}$.

Step 2. For $t \ge 1$, G contains a K_k, S: if G is complete, $n \ge k$; otherwise adding a missing edge creates a K_{k+1}, and its other $k-1$ vertices plus one endpoint form a K_k in G.

Step 3. S has $\binom{k}{2}$ edges, each outside vertex has $\le k-1$ neighbours in S, and $G - S$ on $(t-1)k + r$ vertices has $\le T(n-k,k)$ edges by induction:

$$e(G) \le \binom{k}{2} + (n-k)(k-1) + T(n-k,k) = T(n,k). \tag{11.2}$$

Step 4 (the identity (11.2)) [Supplement]. Removing one vertex from each class of H leaves the Turán graph on $n-k$ vertices; the removed edges are the $\binom{k}{2}$ among removed vertices and $k-1$ per remaining vertex. ∎

Odd cycles of length ≥ 5 have $\chi = 3$ without K_3; odd cycles and complete graphs are the only obstacles:

Theorem 11.19 (Brooks)

If G is connected, neither a complete graph nor an odd cycle, and all degrees are at most d with $d \ge 3$, then $\chi(G) \le d$.

Proof [Supplement] (Lovász).

Step 1 (greedy towards a root). If $\deg w < d$, colour greedily from $[d]$ in non-increasing distance from w. Each $v \ne w$ has a later-coloured neighbour, so $\le d-1$ coloured neighbours; w has $< d$ neighbours. So assume G is d-regular.

Step 2 (a cut vertex). If v is a cut vertex with components C_i of $G - v$, each $G_i = G[C_i \cup \{v\}]$ has $\deg v < d$ and is d-colourable by Step 1 (or complete on $\le d$ vertices). Permute colours to agree at v and combine.

Step 3 (the key triple). Let G be 2-connected; find w with non-adjacent neighbours u, v and $G - u - v$ connected.

If G is 3-connected: G is not complete, so some w on a shortest path between non-adjacent vertices has non-adjacent neighbours u, v, and $G - u - v$ is connected.

If $\{w, y\}$ is a 2-vertex cut: $G - w$ has cut vertex y, hence ≥ 2 leaf blocks, each with a non-cut vertex adjacent to w (else G has a cut vertex). Take such u, v in different leaf blocks (non-adjacent). Degrees in $G - w$ are $\ge d - 1 \ge 2$, so leaf blocks are not single edges and $G - w - u - v$ is connected; w keeps a neighbour there ($d \ge 3$), so $G - u - v$ is connected.

Step 4 (colouring). Colour u, v with 1; colour the rest greedily from $[d]$ in non-increasing distance from w in $G-u-v$, ending with w. Each $v' \neq w$ has $\leq d-1$ coloured neighbours; the d neighbours of w use $\leq d-1$ colours since u, v agree. ■

For every n there are triangle-free graphs with $\chi = n$.

11.5 Matchings in graphs that are not bipartite

Motivation. Tutte: deleting s vertices leaves at most s odd components; sufficiency via saturated graphs.

$c_o(G-S)$ is the number of odd components of $G-S$.

Theorem 11.20 (Tutte)

G has a perfect matching iff $c_o(G-S) \leq |S|$ for every vertex set S.

Only if. Each odd component of $G-S$ has a vertex matched into S, to distinct partners.

G is **saturated non-factorizable** if it has no perfect matching but adding any edge creates one.

Lemma 11.21

If G is saturated non-factorizable and S is the set of vertices adjacent to all other vertices, then the components of $G-S$ are complete graphs.

Proof. Let $ab, bc \in E(G-S)$, $ac \notin E$, and $d \not\sim b$ ($b \notin S$). Perfect matchings $F_1 \ni ac$ of $G+ac$ and $F_2 \ni bd$ of $G+bd$ give $F_1 \triangle F_2$, a union of even alternating cycles; let $C_1 \ni ac$, $C_2 \ni bd$.

Step (a) $C_1 \neq C_2$. $F_1 \triangle C_1$ is a perfect matching of G, a contradiction.

Step (b) $C_1 = C_2$. Traverse C_1 from b through d to the first of a, c, say a, along P. Then $F_2 \triangle (P \cup ab)$ is a perfect matching of G avoiding bd, a contradiction. ■

Theorem 11.22

G is saturated non-factorizable iff either (a) G is complete with an odd number of vertices, or (b) G has an even number of vertices and consists of vertex-disjoint complete graphs $S_0, G_1, \ldots, G_k$ with $k = |S_0| + 2$, each $|G_i|$ odd, and every vertex of every G_i joined to every vertex of S_0.

Proof.

Step 1 (odd order). Odd graphs have no perfect matching, and only K_n admits no new edge.

Step 2 (even order: structure). With $S_0 = S$ as in Lemma 11.21, S_0 and the G_i are complete and S_0 is joined to all. If $c_o(G-S) \leq |S|$: match one vertex of each odd G_i into S, the rest of each G_i internally, the (even) rest of S among itself, a contradiction. So $c_o \geq |S|+2$ by parity. If $c_o > |S|+2$ or an even component existed, adding an edge between two components keeps $c_o(G'-S) > |S|$, contradicting saturation. So $k = |S_0| + 2$, all odd.

Step 3 (conversely) [Supplement]. In (b), $c_o(G-S_0) = |S_0|+2 > |S_0|$: no perfect matching. A missing edge $g_i g_j$ joins two G_i; using it, the other $|S_0|$ odd components each send one vertex into S_0 and the rest match inside: a perfect matching. ■

Proof of Theorem 11.20 ("if", Hetyei–Lovász). Let G satisfy Tutte's condition without a perfect matching; $S = \emptyset$ gives $|V(G)|$ even. Saturate to $G' \supseteq G$ as in Theorem 11.22(b), with

universal set S' and $|S'|+2$ odd complete components of $G'-S'$. Each contains an odd component of $G-S'$, so $c_o(G-S') \geq |S'|+2 > |S'|$, a contradiction. ■

11.6 Selected exercises

Motivation. Hall's condition via a single double count.

Exercise 11.4 (A degree condition for Hall)

Let G be a bipartite graph with colour classes X, Y in which every vertex of X has degree at least as large as every vertex of Y, and every vertex of X has at least one neighbour. Prove that X can be matched into Y.

Solution. Let $a = \min_{x\in X} \deg x \geq b = \max_{y\in Y} \deg y$, $a \geq 1$.

Step 1 (double count). For $T \subseteq X$, the edges E_T at T satisfy $a|T| \leq |E_T| \leq b\,|N(T)|$.

Step 2 (Hall's condition). So $a|T| \leq a|N(T)|$, $|T| \leq |N(T)|$; apply Theorem 11.13.

$a \geq 1$ only excludes the edgeless case $a = b = 0$. ■

Exercise 11.9 (Magic squares are sums of permutation matrices)

Let A be an $n \times n$ matrix with nonnegative integer entries in which every row and every column has the same sum r. Prove that A is a sum of r permutation matrices.

Solution. Induction on r; $r = 0$ is the empty sum. Let $r \geq 1$.

Step 1 (a bipartite graph). Rows $X = \{x_i\}$, columns $Y = \{y_j\}$, $x_iy_j \in E$ iff $a_{ij} > 0$.

Step 2 (Hall's condition). For $T \subseteq X$, the rows of T sum to $r|T|$, all their nonzero entries lie in columns of $N(T)$, which sum to $r|N(T)|$; so $|T| \leq |N(T)|$.

Step 3 (one permutation matrix). Theorem 11.13 gives a perfect matching $x_iy_{\sigma(i)}$, so $a_{i\sigma(i)} \geq 1$; let P_σ be its permutation matrix.

Step 4 (induction). $A - P_\sigma \geq 0$ has line sums $r-1$, so it is a sum of $r-1$ permutation matrices. ■

Chapter 12. Planar Graphs

12.1 Euler's theorem for planar graphs

Motivation. Euler's formula $V + F = E + 2$ plus edge counts around faces rules out $K_{3,3}$ and K_5.

Definition 12.1 (Planar graph)

A graph is **planar** if it can be drawn in the plane with no two edges crossing. The edges of such a drawing cut the plane into regions, the **faces**; one of them, the outer face, is unbounded.

Theorem 12.2 (Euler's theorem on planar graphs)

A connected planar graph with V vertices, E edges and F faces satisfies $V + F = E + 2$.

Proof. Induction on E. $E = 1$: one edge ($V = 2$, $F = 1$) or one loop ($V = 1$, $F = 2$). Let $E \geq 2$.

Step 1 (an edge on a cycle). If $G' = G - e$ is connected, e lies on a cycle, so the two faces beside e merge: G' has V, $E - 1$, $F - 1$, and $V + F - 1 = E - 1 + 2$ by induction.

Step 2 (a tree). Otherwise G is a tree: $V = E + 1$ (Theorem 10.4), $F = 1$. ■

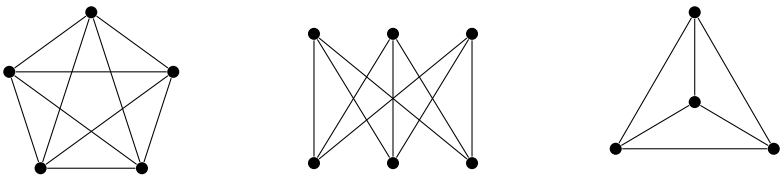
Example 12.3

$K_{3,3}$ is not planar, so the three houses, three wells problem has no solution.
Solution. $V = 6$, $E = 9$ would give $F = 5$. $K_{3,3}$ is simple bipartite, so each face has ≥ 4 edges and $4F \leq 2E$: $20 \leq 18$, a contradiction.

Hence $K_6 \supseteq K_{3,3}$ is not planar; K_4 is.

Example 12.4

K_5 is not planar.
Solution. $V = 5$, $E = 10$ give $F = 7$, and $3F \leq 2E$: $21 \leq 20$, a contradiction.



K_5 $K_{3,3}$ K_4, drawn without crossings

Figure. *The two minimal non-planar graphs and a plane drawing of K_4.*

Subdividing an edge or suppressing a degree-2 vertex preserves planarity; graphs related by these moves are **edge-equivalent** (homeomorphic).

Theorem 12.5 (Kuratowski)

A graph is not planar iff it contains a subgraph that is edge-equivalent to K_5 or $K_{3,3}$.

Proof. "If": such a subgraph is non-planar (Examples 12.3, 12.4), hence so is the graph. The book states "only if" without proof. ■

12.2 Polyhedra

Motivation. Euler's formula leaves room for only five regular polyhedra.

For a convex polyhedron (its 1-skeleton), faces have ≥ 3 edges and vertices degree ≥ 3. It is **regular** if every face is an ℓ-cycle and every vertex has degree d.

Theorem 12.6

A convex polyhedron with V vertices, F faces and E edges satisfies $V + F = E + 2$.

Proof (angle counting). Project P onto a plane perpendicular to no face. Faces project to convex polygons; the outer boundary B is a convex v_1-gon, and $v_2 = V - v_1$ vertices project inside. Each interior point is covered by one front and one back face. Let S be the sum of all angles of projected faces.

Step 1 (by vertices). An inside vertex contributes $360°$; on B, front and back faces each fill the interior angle, contributing twice $(v_1 - 2)180°$:

$$S = (v_1 - 2)360 + 360v_2 = (V - 2)360. \tag{12.1}$$

Step 2 (by faces). With f_i the edge numbers of the faces,

$$\sum_{i=1}^{F} f_i = 2E, \tag{12.2}$$

and a convex k-gon has angle sum $(k - 2)180°$, so

$$S = \sum_{i=1}^{F}(f_i - 2)180 = 180\sum_{i=1}^{F} f_i - 360F = 360(E - F). \tag{12.3}$$

(12.1) and (12.3) give $V - 2 = E - F$. ■

Corollary 12.7

In a convex polyhedron, $3F \leq 2E$.

Proof. $f_i \geq 3$ in (12.2). ■

Proposition 12.8

In a convex polyhedron, $3V \leq 2E$.

Proof. With vertex degrees $c_i \geq 3$,

$$\sum_{i=1}^{V} c_i = 2E. \tag{12.4}$$

■

Lemma 12.9

In a convex polyhedron, $E \leq 3V - 6$ and $E \leq 3F - 6$.

Proof. Corollary 12.7: $E + 2 = V + F \leq V + \frac{2E}{3}$, so $E \leq 3V - 6$. Proposition 12.8: $E + 2 \leq F + \frac{2E}{3}$, so $E \leq 3F - 6$. ■

Remark

Remark[$E \le 3V-6$ for every simple planar graph [Supplement]] For simple connected planar G with $V \ge 3$ (add edges between components): a tree has $E = V-1 \le 3V-6$; otherwise faces have closed boundary walks of length ≥ 3, so $3F \le 2E$ and Euler gives $E \le 3V-6$. Hence some vertex has degree ≤ 5 (else $2E \ge 6V$).

Lemma 12.10

Every convex polyhedron has a face with at most five edges.

Proof. By (12.2) and Lemma 12.9,

$$\sum_{i=1}^{F} f_i = 2E \le 6F - 12 < 6F. \tag{12.5}$$

■

Lemma 12.11

Every convex polyhedron has a vertex of degree at most five.

Proof. By (12.4) and Lemma 12.9,

$$\sum_{i=1}^{V} c_i = 2E \le 6V - 12 < 6V. \tag{12.6}$$

■

Put a vertex in each face of a plane G and join the two faces beside each edge e by an edge crossing e: G^* is planar with F vertices, E edges, V faces.

Definition 12.12 (Dual graph)

G^* is the **dual graph** of the plane graph G.

The dual of a regular polyhedron of type (d, ℓ) is regular of type (ℓ, d).

Classification. Lemmas 12.10, 12.11 give $d, \ell \in \{3,4,5\}$, and by (12.2)

$$E = \frac{F\ell}{2}. \tag{12.7}$$

(A) $d=3$: $3V = 2E$, $3F = E+6$, $(6-\ell)F = 12$; $\ell = 3,4,5$ give $(V,E,F) = (4,6,4)$ (tetrahedron), $(8,12,6)$ (cube), $(20,30,12)$ (dodecahedron).

(B) $d=4$: $E = 2F-4$, $(4-\ell)F = 8$; $\ell = 3$: $(6,12,8)$, octahedron.

(C) $d=5$: $3E = 5F-10$, $(10-3\ell)F = 20$; $\ell = 3$: $(12,30,20)$, icosahedron.

Theorem 12.13

There are exactly five regular polyhedra: the tetrahedron, the cube, the dodecahedron, the octahedron and the icosahedron.

Proof. (A)–(C) leave exactly these five parameter sets, each realised by the named solid. ■

	d	ℓ	V	E	F
Tetrahedron	3	3	4	6	4
Cube	3	4	8	12	6
Octahedron	4	3	6	12	8
Dodecahedron	3	5	20	30	12
Icosahedron	5	3	12	30	20

12.3 Colouring maps

Motivation. Map colouring is vertex colouring of the dual; Kempe chains give five colours.

Colouring faces so that neighbours differ is properly colouring G^*. K_4 shows four colours are needed.

Proposition 12.14

Every planar graph can be properly vertex-coloured with six colours.

Proof. Induction on V, G simple. By the Remark after Lemma 12.9 some A has degree ≤ 5; colour $G - A$ by induction and give A a colour missing from its neighbours. ■

Theorem 12.15 (Five colour theorem)

Every planar graph can be properly vertex-coloured with five colours.

Proof. Induction on V, G simple. Take $\deg A \leq 5$ and colour $G' = G - A$. If A's neighbours use ≤ 4 colours, done. Otherwise they are $y_1, \ldots, y_5$ clockwise, coloured $1, \ldots, 5$.

Step 1 (Kempe chains). Let K be the component of y_1 in the subgraph induced by colours $1, 3$. Swapping $1, 3$ on K keeps the colouring proper; if $y_3 \notin K$, colour 1 becomes free at A.

Step 2 (both chains cannot exist). If $y_3 \in K$, a 1–3 path $y_1 \to y_3$ and A form a cycle C separating y_2 from y_4. Likewise we are done unless a 2–4 path $y_2 \to y_4$ exists; it would meet $C - A$ (coloured $1, 3$) at a vertex, impossible. ■

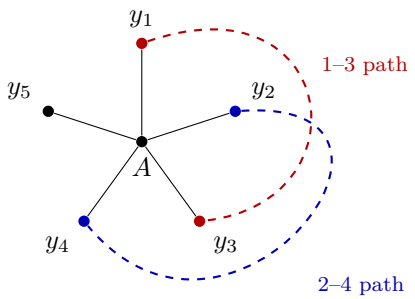


Figure. *Kempe chains: a 1–3 path $y_1 \to y_3$ and a 2–4 path $y_2 \to y_4$ would cross at a vertex.*

Remark

Remark[The four colour theorem] Four colours suffice: Appel–Haken (1976, computer check of 1936 configurations; 633 by Robertson–Sanders–Seymour–Thomas, 1996); machine-verified

by Gonthier–Werner (2005).

12.4 Selected exercises

Motivation. Euler's formula plus two double counts gives one linear relation.

The *length* of a face counts boundary edges (twice if the face lies on both sides); face lengths and degrees each sum to $2E$.

Exercise 12.15 (Polyhedra without triangles)

Let P be a convex polyhedron with no triangular faces. Prove that $E \le 2V - 4$.

Solution.

Step 1 (faces). $4F \le 2E$, so $F \le \frac{E}{2}$.

Step 2 (Euler). By Theorem 12.6, $E + 2 = V + F \le V + \frac{E}{2}$, so $E \le 2V - 4$.

Equality iff all faces are quadrilaterals (cube: $V = 8$, $E = 12$). ■

Exercise 12.19 (Trivalent (a, b)-faced polyhedra)

Let P be a convex polyhedron whose faces are all a-gons or b-gons and in which every vertex lies on exactly three edges. Let p_a, p_b be the numbers of a-gonal and b-gonal faces and n the number of vertices.
(a) Express the number E of edges in two ways.
(b) Prove that $p_a(6 - a) + p_b(6 - b) = 12$.

Solution.

(a) $2E = a\,p_a + b\,p_b$ (faces) and $2E = 3n$ (vertices).

(b) Multiply $n + p_a + p_b = E + 2$ (Theorem 12.6) by 6 and use $6n = 4E$:

$$4E + 6p_a + 6p_b = 6E + 12, \qquad \text{so} \qquad 6p_a + 6p_b - 2E = 12,$$

and substitute $2E = a\,p_a + b\,p_b$.

E.g. a fullerene (pentagons and hexagons) has exactly 12 pentagons. ■

Exercise 12.22 (Octahedrites)

Let G be a connected planar graph in which each face has length 2, 3 or 4, with p_2, p_3, p_4 faces of these lengths. Assume that every vertex has degree 4 and that $p_2 + p_3 = 8$. Prove that $p_2 = 0$ and $p_3 = 8$.

Solution.

Step 1 (two counts). $V = \frac{E}{2}$, $2E = 2p_2 + 3p_3 + 4p_4$, $F = p_2 + p_3 + p_4$.

Step 2 (Euler). Theorem 12.2 gives $F = \frac{E}{2} + 2$; times 4:

$$4p_2 + 4p_3 + 4p_4 = 2E + 8 = 2p_2 + 3p_3 + 4p_4 + 8, \qquad \text{so} \qquad 2p_2 + p_3 = 8.$$

Step 3 (conclusion). Subtracting $p_2 + p_3 = 8$: $p_2 = 0$, $p_3 = 8$. ■

Chapter 13. Ramsey Theory

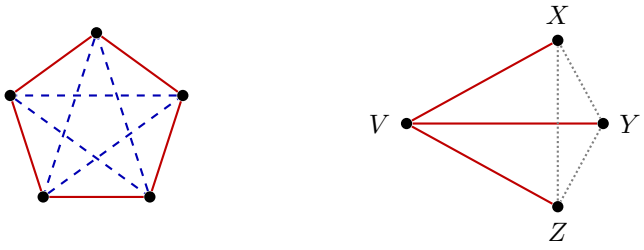
13.1 Ramsey theory for finite graphs

Motivation. Large 2-coloured cliques contain large monochromatic ones, via $R(k,\ell) \le R(k,\ell-1) + R(k-1,\ell)$.

Example 13.1

Among any six people there are three who all know each other or three who are mutual strangers.
Solution. Colour K_6 red (acquainted) / blue. A vertex V has 5 edges, so three of the same colour, say red, go to X, Y, Z. A red edge in XYZ gives a red triangle with V; otherwise XYZ is blue.

Five are not enough: pentagon sides red, diagonals blue.



K_5: no monochromatic triangle Example 13.1

Figure. *Left: K_5 without monochromatic triangles. Right: a red edge in XYZ closes a red triangle.*

Equivalently, every 6-vertex simple H or its complement contains a **clique** of size 3.

Theorem 13.2 (Ramsey's theorem for graphs)

For integers $k, \ell \ge 2$ there is a least positive integer $R(k,\ell)$ such that every red/blue colouring of the edges of $K_{R(k,\ell)}$ contains a red K_k or a blue K_ℓ.

(One such integer suffices; then a least exists.)

Example 13.3

$R(3,3) = 6$ by Example 13.1 and the pentagon. Trivially $R(2,2) = 2$.

Proof of Theorem 13.2. Induction on $k + \ell$.

Step 1 (boundary). $R(k,2) = k$: K_k is all red or has a blue K_2; likewise $R(2,\ell) = \ell$.

Step 2 (induction step). We show

$$R(k,\ell) \le R(k,\ell-1) + R(k-1,\ell). \tag{13.1}$$

In K_N, N the right side, a vertex V has $N-1$ edges: $\ge R(k,\ell-1)$ blue or $\ge R(k-1,\ell)$ red.

(a) Blue neighbours contain a red K_k or a blue $K_{\ell-1}$, which with V is a blue K_ℓ.

(b) Red neighbours contain a blue K_ℓ or a red K_{k-1}, which with V is a red K_k. ∎

Corollary 13.4

For $k, \ell \geq 2$,

$$R(k,\ell) \leq R(k,\ell-1) + R(k-1,\ell). \tag{13.2}$$

Proof. This is (13.1). ■

E.g. $R(4,3) \leq R(4,2) + R(3,3) = 10$, not sharp:

Example 13.5

$R(4,3) = 9$.
Solution.
(1) K_9 *always works.* Some V has (i) ≥ 6 red or (ii) ≥ 4 blue edges; otherwise all red degrees are 5, with odd sum 45.
(a) Six red neighbours span a triangle (Example 13.1): blue, or red giving a red K_4 with V.
(b) Four blue neighbours: all edges red give a red K_4; a blue edge gives a blue triangle with V.
(2) K_8 *need not work.* On $1,\ldots,8$ colour $\{i,j\}$ $(i<j)$ blue if $j-i \in \{1,4,7\}$, red if $j-i \in \{2,3,5,6\}$. A blue triangle with least vertex i needs two of $i+1, i+4, i+7$ (differences $3,6,3$: red). A red K_4 needs three of $i+2, i+3, i+5, i+6$, hence a pair at difference 1 (blue).

Example 13.6

$R(4,4) = 18$.
Solution. $R(4,4) \leq R(4,3) + R(3,4) = 18$. On $\mathbb{Z}_{17}$, colour $\{i,j\}$ red iff $i-j \in Q = \{\pm1, \pm2, \pm4, \pm8\}$, the nonzero squares (well defined, $-1 \in Q$).
[Supplement] *No monochromatic* K_4. Translations and multiplications by $q \in Q$ preserve the red graph, so a red K_4 normalises to $\{0,1,a,b\}$ with $a,b \in Q \cap (1+Q) \setminus \{0,1\} = \{2,9,16\}$; but differences of $2, 9, 16$ are $\pm7, \pm14 \notin Q$. Multiplication by the non-square 3 swaps the colours, so there is no blue K_4 either.

$R(4,5) = 25$, $43 \leq R(5,5) \leq 48$ (as quoted in the book); $R(k,k)$ is unknown for $k \geq 5$.

Theorem 13.7

For $k, \ell \geq 2$,

$$R(k,\ell) \leq \binom{k+\ell-2}{k-1}. \tag{13.3}$$

Proof. Induction on $k+\ell$; for $k=2$ or $\ell=2$ both sides are equal. Otherwise, by Corollary 13.4, induction and Pascal,

$$R(k,\ell) \leq \binom{k+\ell-3}{k-1} + \binom{k+\ell-3}{k-2} = \binom{k+\ell-2}{k-1}.$$

■

Corollary 13.8

$R(k,k) \leq 4^{k-1}$ for $k \geq 2$.

Proof. $R(k,k) \leq \binom{2k-2}{k-1} \leq \sum_i \binom{2k-2}{i} = 2^{2k-2}$. ■

Campos–Griffiths–Morris–Sahasrabudhe (2023): $R(k,k) < (4-\epsilon)^k$. Exponential lower bounds: Chapter 15.

13.2 Generalizations of the Ramsey theorem

Motivation. The same induction handles many colours and r-subsets; a corollary is Erdős–Szekeres.

Example 13.9

If every edge of K_{17} is coloured with one of three colours, there is a monochromatic triangle. *Solution.* V has 16 edges, six of one colour, say green, to a set g. A green edge in g closes a green triangle; otherwise g is 2-coloured on ≥ 6 vertices (Example 13.1).

Theorem 13.10

For positive integers $n_1,\dots,n_k$ there is a least $N = R(n_1,\dots,n_k)$ such that every colouring of the edges of K_n, $n \geq N$, with colours $1,\dots,k$ has, for some i, a K_{n_i} all of whose edges have colour i.

Proof (completed [Supplement]). Induction on $\sum n_i$. If some $n_i = 1$, $N = 1$. Otherwise with $R_i = R(n_1,\dots,n_i-1,\dots,n_k)$ let

$$N = \sum_{i=1}^{k}(R_i - 1) + 2.$$

A vertex V has $\sum_i (R_i-1)+1$ edges, so some colour i occurs on $\geq R_i$ of them, ending in S. Then K_S has a K_{n_j} of colour $j \neq i$, or a K_{n_i-1} of colour i which with V is a K_{n_i}. ■

Colouring r-subsets instead of edges ($r = 2$):

Theorem 13.11 (Hypergraph Ramsey theorem)

Colour each r-element subset of an n-set with one of k colours, and let $n_1,\dots,n_k$ be positive integers. There is a least $N = R_r(n_1,\dots,n_k)$ such that for $n \geq N$ there are an index i and an n_i-set all of whose r-subsets have colour i.

Proof [Supplement]. Induction on r, then on $\sum n_i$.

Step 1 ($r = 1$). $N = \sum_i (n_i - 1) + 1$ by pigeon-hole.

Step 2 (trivial cases). If some $n_i < r$, any n_i-set qualifies vacuously: $N = n_i$.

Step 3 (induction step). Let $r \geq 2$, all $n_i \geq r$, $R_i = R_r(n_1,\dots,n_i-1,\dots,n_k)$ and

$$N = 1 + R_{r-1}(R_1,\dots,R_k).$$

Fix $V \in X$, $|X| = N$, and colour each $(r-1)$-set $S \subseteq X - V$ by the colour of $S \cup \{V\}$. Some $T \subseteq X - V$, $|T| = R_i$, has all $S \cup \{V\}$ ($S \subseteq T$) of colour i. On T, either some n_j-set ($j \neq i$) is monochromatic of colour j, or some (n_i-1)-set U has all r-subsets of colour i; then so does $U \cup \{V\}$. ■

Theorem 13.12 (Erdős–Szekeres)

For every n there is a least $ES(n)$ such that among any $N \geq ES(n)$ points in the plane, no three collinear, some n form a convex n-gon.

Proof. $R_3(n,n)$ points suffice. Number the points; colour a triangle red if its vertices in increasing order turn clockwise, blue otherwise. Theorem 13.11 gives n points with all triangles of one colour.

Step 1 (reduction) [Supplement]. n points are not in convex position iff one lies inside a triangle of three others: a non-hull point P lies in a triangle of a fan triangulation of the hull, not on its boundary (no three collinear).

Step 2 (no point inside a triangle). Let $\chi(x,y,z) = \pm 1$ be the orientation: cyclic-invariant, sign-changing under transpositions; a triangle $x<y<z$ has colour $\chi(x,y,z)$. If D is inside ABC,

$$\chi(A,B,D) = \chi(B,C,D) = \chi(C,A,D) = \chi(A,B,C) =: \sigma.$$

By a cyclic shift and a reflection (reversing all orientations) take $A<B<C$, all four triangles of colour σ. Then:

(a) If $D<A$: $\chi(D,A,C) = \chi(A,C,D) = -\chi(C,A,D) = -\sigma$.

(b) If $A<D<B$: $\chi(A,D,B) = -\chi(A,B,D) = -\sigma$.

(c) If $B<D<C$: $\chi(B,D,C) = -\chi(B,C,D) = -\sigma$.

(d) If $C<D$: $\chi(A,C,D) = -\sigma$.

Each gives a triangle of colour $-\sigma$, a contradiction. ∎

Erdős–Szekeres: $2^{n-2}+1 \le ES(n) \le \binom{2n-4}{n-2}+1$, equality on the left conjectured (known for $n \le 6$); Suk (2017): $ES(n) \le 2^{n+o(n)}$.

13.3 Ramsey theory in geometry

Motivation. Colourings of the plane force monochromatic configurations.

Example 13.13

If every point of the plane is red or blue, some two points at unit distance have the same colour.
Solution. Two vertices of a unit equilateral triangle agree.

Example 13.14

The same holds with three colours.
Solution. Otherwise every unit equilateral triangle ABC is rainbow. Reflecting A across BC gives D with BCD unit equilateral, so D has A's colour and $|AD| = \sqrt{3}$. Hence the circle k of radius $\sqrt{3}$ about a red point is red; but k contains two points at distance 1, a contradiction.

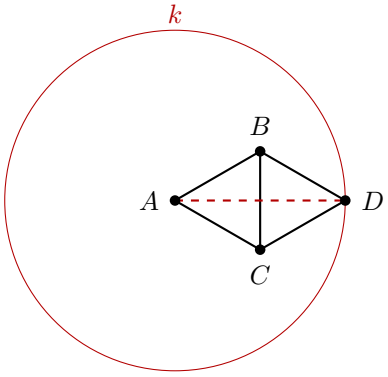


Figure. *Example 13.14: D takes A's colour, hence so does all of k.*

Example 13.15

Colour the plane red/blue, and let T be a triangle with angles $30°, 60°, 90°$ and hypotenuse 1. Some triangle congruent to T has monochromatic vertices.
Solution. Take a red unit segment AB (Example 13.13) and $D_1, \dots, D_4$ dividing the circle on diameter AB, with A, B, into six equal arcs; each $ABD_i \cong T$ (Thales). If all D_i are blue, two opposite D_i and a third form a blue copy of T.

So the **chromatic number of the plane** (of its unit-distance graph) exceeds 3; it is at most 7, and ≥ 5 (de Grey, 2018).

13.4 Selected exercises

Motivation. Pigeon-hole at one vertex, then induction, in three settings.

Exercise 13.2 (Monotone subsequences)

Prove that every permutation of length $nm+1$ has an increasing subsequence of length $n+1$ or a decreasing subsequence of length $m+1$. (The entries of a subsequence need not be in consecutive positions.)

Solution.

Step 1 (labels). Let $\lambda(x)$ be the length of the longest increasing subsequence ending at x; assume $\lambda(x) \in [n]$ for all x.

Step 2 (pigeon-hole). $nm+1$ entries, n labels: $\geq m+1$ share a label (Theorem 1.4).

Step 3 (equal labels decrease). If x precedes y with $\lambda(x) = \lambda(y)$ and $x < y$, then $\lambda(y) \geq \lambda(x) + 1$. So equal-label entries form a decreasing subsequence of length $\geq m+1$.

Sharp: $2\,1\,4\,3$ ($n = m = 2$). ■

Exercise 13.6 (Monochromatic triangles in k colours)

Let $n_k = k!(1 + \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{k!}) + 1$. Prove that every colouring of the edges of K_{n_k} with k colours contains a triangle with monochromatic edges.

Solution. Let $m_k = n_k - 1 = \sum_{j=0}^{k} \frac{k!}{j!}$. For $k \geq 2$,

$$m_k = \sum_{j=0}^{k-1} k \cdot \frac{(k-1)!}{j!} + \frac{k!}{k!} = k\, m_{k-1} + 1.$$

Induction on k: every k-colouring of K_{m_k+1} has a monochromatic triangle.

Step 1 (initial step). $m_1 = 2$: K_3 in one colour.

Step 2 (pigeon-hole at a vertex). A vertex v has $k\, m_{k-1} + 1$ edges, so some colour, say 1, is on $\geq m_{k-1} + 1$ of them, ending in S (Theorem 1.4).

Step 3 (inside S). An edge of colour 1 in S closes a triangle with v; otherwise S uses $k-1$ colours and induction applies.

$k = 2, 3$ give $n_2 = 6$ and $n_3 = 17$ (Example 13.9). ■

Exercise 13.16 (Schur's theorem)

Colour every positive integer with one of k colours. Prove that there is an integer $N = N(k)$ such that for every $n > N$ there are integers $a, b, c < n$ of the same colour with $a + b = c$ (with $a = b$ allowed).

Solution.

Step 1 (the constant). $N = R(3, \dots, 3)$ (k entries, Theorem 13.10); let $n > N$.

Step 2 (an edge colouring). On $0, \dots, N-1$ give $\{i, j\}$ the colour of $|i - j| \in [N-1]$.

Step 3 (a monochromatic triangle). Some $i < j < l$ has one colour on its edges: $a = j - i$, $b = l - j$, $c = l - i$ agree in colour, $a + b = c$, and $a, b, c \le N - 1 < n$. ■

Chapter 14. Subsequence Conditions on Permutations

14.1 Pattern avoidance

Motivation. Every length-3 pattern is avoided by c_n permutations; length 4 splits into three classes.

A lineup of children is good (each sees every shorter child ahead) iff there are no entries a, b, c, in this order, with $a < c < b$: 6723415 is good, 1423567 is not (4 blocks 1 from 2).

Definition 14.1 (132-pattern)

Entries a, b, c of a permutation, in this left-to-right order, with $a < c < b$ form a 132**-pattern** (they are ordered like $1, 3, 2$). Similarly for 123, 231, etc.

Definition 14.2 (132-avoiding)

A permutation with no 132-pattern is 132**-avoiding**.

Let $f(n)$ count 132-avoiding n-permutations ($f(0) = 1$). With n in position i, every entry left of n exceeds every entry right of it (else $x\,n\,y$ is a 132), so the sides are 132-avoiders of $\{n-i+1, \dots, n-1\}$ and $\{1, \dots, n-i\}$, and conversely. Hence

$$f(n) = \sum_{i=1}^{n} f(i-1) f(n-i) \qquad (n \geq 1). \tag{14.1}$$

Corollary 14.3

The number of 132-avoiding n-permutations is the Catalan number $c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. (14.1) with $f(0) = 1$ is the Catalan recurrence $c_n = \sum_{i=1}^{n} c_{i-1} c_{n-i}$ (from (8.6), §8.1.2.1); induct. ∎

Definition 14.4 (Pattern)

Let p be an n-permutation and $q = q_1 \cdots q_k$ a k-permutation, $k \leq n$. Entries $a_1, \dots, a_k$ of p (left to right) form a q**-pattern** if $a_i < a_j \iff q_i < q_j$.

Definition 14.5 (Avoidance)

p is q**-avoiding** if no k entries of p form a q-pattern. $S_n(q)$ (or $\mathrm{Av}_n(q)$) denotes the number of q-avoiding n-permutations.

Symmetries. The **reverse** $p_n \cdots p_1$ and the **complement** $\bar{p} = (n+1-p_1) \cdots (n+1-p_n)$ biject q-avoiders onto avoiders of the reversed/complemented pattern, so

$$S_n(132) = S_n(231) = S_n(312) = S_n(213), \qquad S_n(123) = S_n(321).$$

A **left-to-right minimum** is an entry smaller than all before it; these form a decreasing subsequence from p_1 to 1.

Lemma 14.6

$S_n(123) = S_n(132)$ for all $n \geq 1$.

Proof. A bijection from 123- to 132-avoiders fixing the left-to-right minima (values and positions).

Step 1 (constraints). Let $m_1 > \cdots > m_r$ be the minima at positions $1 = i_1 < \cdots < i_r$. The other positions are *slots*; a slot between i_j and i_{j+1} has *threshold* m_j. A filling keeps exactly these minima iff each slot value exceeds its threshold. Thresholds decrease.

Step 2 (the map f). For a 123-avoider p, fill slots left to right with the smallest unused value above the threshold. At the s-th slot (threshold m), the first s slots of p hold s values $> m$ and only $s-1$ are used, so this never gets stuck.

Step 3 ($f(p)$ avoids 132). Non-minima x before y, both above a minimum m_j to their left, satisfy $x < y$ (y was available when x was placed). A 132-pattern a, b, c may take $a = m_j$ (the last minimum at or before a); then $b, c > m_j$ are non-minima to its right, so $b < c$, a contradiction.

Step 4 (the inverse g). For a 132-avoider q, fill the slots with the non-minima in *decreasing* order; this respects thresholds (the first s slots of q hold s values above the s-th threshold). $g(q)$ is a union of two decreasing sequences, so avoids 123.

Step 5 (g and f are inverse). In a 123-avoider the non-minima decrease (else the minimum before x, x, y is a 123), so $g(f(p)) = p$. A 132-avoider q is the greedy filling: if a slot (threshold m_j) holds x while a smaller eligible $y > m_j$ comes later, m_j, x, y is a 132. So $f(g(q)) = q$. ■

E.g. $p = 465132$ has minima $4, 1$ and $f(p) = 456123$.

Theorem 14.7

For every pattern q of length 3 and all $n \geq 1$, $S_n(q) = c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. Symmetries, Lemma 14.6 and Corollary 14.3. ■

Length 4 leaves three classes 1342, 1234, 1324:

$$\begin{array}{ll} S_n(1342): & 1, 2, 6, 23, 103, 512, 2740, 15485 \\ S_n(1234): & 1, 2, 6, 23, 103, 513, 2761, 15767 \\ S_n(1324): & 1, 2, 6, 23, 103, 513, 2762, 15793 \end{array}$$

($n \leq 8$). In fact $S_n(1342) < S_n(1234) < S_n(1324)$ for $n \geq 7$; we prove the second inequality.

Theorem 14.8

$S_n(1234) < S_n(1324)$ for all $n \geq 7$.

A **right-to-left maximum** exceeds all entries to its right. Proof after Lemma 14.11.

Definition 14.9 (Class)

Permutations x, y are in the same **class** if they have the same left-to-right minima at the same positions and the same right-to-left maxima at the same positions.

E.g. 51234, 51324 share a class; 24315, 24135 do not.

Constraints of a class [Supplement]. Each remaining position (slot) s has a lower bound $l(s)$ (nearest minimum to its left) and upper bound $u(s)$ (nearest maximum to its right); a filling lies in the class iff $l(s) <$ value $< u(s)$ for all slots. l, u are non-increasing.

Lemma 14.10

Every nonempty class contains exactly one 1234-avoiding permutation.

Proof.

Step 1 (existence). Fill the slots in decreasing order. [Supplement] In any member, the first s slots hold s values $> l(s)$ and the last $N-s+1$ slots values $< u(s)$, so the s-th largest value lies in $(l(s), u(s))$. The result is a union of three decreasing sequences, so by pigeon-hole has no 1234.

Step 2 (uniqueness). In a 1234-avoider, slot entries $a < b$ in this order with the minimum before a and the maximum after b would form 1234; so slot entries decrease, as in Step 1. ■

An **inversion** of p is a pair $i < j$ with $p_i > p_j$.

Lemma 14.11

Every nonempty class contains at least one 1324-avoiding permutation.

Proof.

Step 1 (good patterns). In a 1324-pattern $a\,c\,b\,d$, replace a by the nearest minimum at or before it and d by the nearest maximum at or after it: a **good pattern**. Its middle entries c, b are slot entries (minima can only play "1", maxima "4").

Step 2 (swapping). Exchange c and b: at its new place $b > a \geq l$ and $b < c < u$; $c < d \leq u$ and $c > b > l$. So the class is kept, and the number of inversions drops.

Step 3 (termination). After at most $\binom{n}{2}$ swaps no good pattern, hence no 1324, remains. ■

Proof of Theorem 14.8. Lemmas 14.10, 14.11 give $S_n(1234) \leq S_n(1324)$. Write $a_1{*}a_2{**}b_1$ for the class with minima a_1, a_2 at positions $1, 3$ and maximum b_1 at position 6. For $n = 7$, $3{*}1{*}7{*}5$ contains the two 1324-avoiders 3612745, 3416725; for $n > 7$ use $n(n-1)\cdots 8\,3{*}1{*}7{*}5$. ■

Stanley–Wilf. $S_n(q) \leq c_q^n$ for every q (Marcus–Tardos, 2004; cited). For monotone patterns:

Theorem 14.12

For $k \leq n$, $S_n(12\cdots k) \leq (k-1)^{2n}$.

Proof. The **rank** of x is the length of the longest increasing subsequence ending at x; equal ranks form decreasing subsequences. A $12\cdots k$-avoider has ranks in $[k-1]$, and is determined by assigning values ($(k-1)^n$ ways) and positions ($(k-1)^n$ ways) to the $k-1$ decreasing subsequences. ■

Compare $S_n(123) = c_n < 4^n$.

Definition 14.13 (Direct sum)

For $p \in S_a$, $q \in S_b$, the **direct sum** $p \oplus q \in S_{a+b}$ is p followed by q with every entry increased by a.

Example 14.14

$132 \oplus 2431 = 1325764$.

Theorem 14.15

If $S_n(q_1 \oplus 1) \le c_1^n$ and $S_n(1 \oplus q_2) \le c_2^n$ for all n, then

$$S_n(q_1 \oplus 1 \oplus q_2) \le \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

Example 14.16

$S_n(2134) = S_n(1234) < 9^n$ (Wilf-equivalence cited; bound by Theorem 14.12), likewise $S_n(1243) < 9^n$. With $q_1 = 213$, $q_2 = 132$: $S_n(213 \oplus 1 \oplus 132) \le (3+3)^{2n} = 36^n$.

Proof of Theorem 14.15. Let p avoid $q = q_1 \oplus 1 \oplus q_2$; colour an entry red if it ends some $(q_1 \oplus 1)$-pattern, blue otherwise.

Step 1. Red entries avoid $1 \oplus q_2$: the first entry x of a red copy ends a $q_1 \oplus 1$ whose other entries are left of and below x, completing a q.

Step 2. Blue entries avoid $q_1 \oplus 1$ (its last entry would be red).

Step 3 (counting). With k blue entries, p is fixed by the blue values, blue positions and the two patterns: $\le \binom{n}{k}^2 c_1^k c_2^{n-k}$. So

$$S_n(q) \le \sum_{k=0}^{n} \left(\binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k} \right)^2 \le \left(\sum_{k=0}^{n} \binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k} \right)^2 = \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

■

14.2 Stack sortable permutations

Motivation. One stack sorts exactly the 231-avoiders; $s(LnR) = s(L)s(R)n$ reveals a descent symmetry.

Stack sorting. Read p left to right; push an entry if it is smaller than the top, otherwise pop the top to the output until it can be pushed; empty the stack at the end. The output is $s(p)$; p is **stack sortable** if $s(p) = 12\cdots n$.

Example 14.17

$p = 2413$: push 2; 4 pops 2, pushed; push 1; 3 pops 1, pushed on 4; pop 3, 4. So $s(2413) = 2134$: not stack sortable.

Proposition 14.18

Let $a < b$ be entries of p.
(1) If a precedes b in p, then a precedes b in $s(p)$.
(2) If b precedes a and no entry $c > b$ lies between them, then a precedes b in $s(p)$.
(3) If b precedes a and some $c > b$ lies between them (so b, c, a form a 231-pattern), then b precedes a in $s(p)$.

Proof.

(1) b cannot go on top of a, so a leaves first.

(2) An entry leaves only when a larger one arrives or at the end; so b is still stacked when a goes above it. [Supplement] (The book's claim that the entries between are decreasing is not needed.)

(3) b leaves before c is pushed, which is before a arrives. ■

Theorem 14.19

p is stack sortable iff it avoids 231.

Proof. A 231 $b\,c\,a$ keeps b before a (part 3); otherwise every pair falls under (1) or (2). ■

So c_n n-permutations are stack sortable. p is **two-stack sortable** if $s(s(p))$ is the identity; this is not closed under patterns:

Example 14.20

$s(35241) = 32145$ and $s(32145) = 12345$, so 35241 is two-stack sortable. But its subsequence 3241 has $s(3241) = 2314$ and $s(2314) = 2134$.

Theorem 14.21 (West)

p is two-stack sortable iff it contains no 2341-pattern and no 3241-pattern except as part of a 35241-pattern.

Proof.

Step 1 (a 2341 *obstructs).* For $b\,c\,d\,a$ $(a < b < c < d)$, b precedes c (part 1) and both precede a (part 3, via d) in $s(p)$: a 231.

Step 2 (a bad 3241 *obstructs).* For $y\,x\,z\,w$ $(w < x < y < z)$ not in a 35241, x, y precede w in $s(p)$ (via z). If nothing between y, x exceeds y, x precedes y (part 2) and x, y, w is a 231. Else some such t has $t < z$ (no 35241), so $y\,t\,z\,w$ is a 2341 (Step 1).

Step 3 (sufficiency). Let $f\,g\,e$ be a 231 in $s(p)$ $(e < f < g)$. By Proposition 14.18, f, g precede e in p with some $h > g$ between g and e.

(a) If f precedes g in p, then $f\,g\,h\,e$ is a 2341.

(b) If g precedes f: nothing between them exceeds g (part 3), so h is between f and e and $g\,f\,h\,e$ is a 3241, not inside a 35241 (which needs an entry $> h$ between g and f). ■

Two-stack sortable n-permutations number $\frac{2}{(n+1)(2n+1)}\binom{3n}{n}$ (cited). p is **t-stack sortable** if $s^t(p)$ is the identity.

Lemma 14.22

If $p = LnR$, then $s(p) = s(L)\,s(R)\,n$.

Proof. L leaves before n enters, giving $s(L)$; R passes over n, giving $s(R)$; n leaves last. ■

Corollary 14.23

Every n-permutation is $(n-1)$-stack sortable.

Proof. Induction on n: $s(\sigma n) = s(\sigma)n$, so $s^{n-1}(p) = s^{n-2}(s(L)s(R))\,n$, with $s(L)s(R)$ a permutation of $[n-1]$. ■

Corollary 14.24

$s^t(p)$ ends with $(n-t+1)(n-t+2)\cdots n$.

Proof. Induction on t [Supplement]: if $s^{t-1}(p) = \sigma\,(n-t+2)\cdots n$, Lemma 14.22 gives $s^t(p) = s(\sigma)\,(n-t+2)\cdots n$, and $s(\sigma)$ ends with $\max\sigma = n-t+1$. ■

Decreasing binary trees. $T(p)$ has root n, left child $\max L$ and right child $\max R$ (an only child is still left or right), with $T(L)$, $T(R)$ below. In-order reading recovers p.

Example 14.25

For $p = 263498175$, $T(p)$ is drawn below.

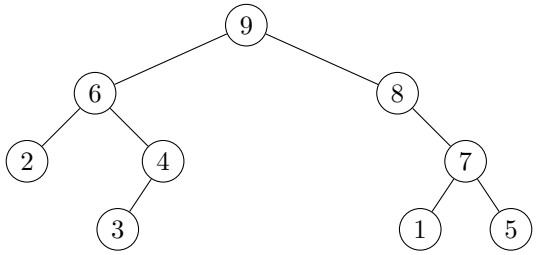


Figure. $T(263498175)$: *in order* 263498175, *postorder* 234615789 $= s(p)$.

Example 14.26

Postorder (left, right, root) of the tree above gives $234615789 = s(263498175)$.

Proposition 14.27

$T(p)$ read in postorder is $s(p)$.

Proof. Induction: postorder of $T(LnR)$ is $s(L)s(R)n = s(p)$ (Lemma 14.22). ■

i is a **descent** if $p_i > p_{i+1}$; $d(p)$ counts them. By complementation, k and $n-1-k$ descents are equally frequent overall. [Supplement] $p_i > p_{i+1}$ iff p_i has a right child (its in-order successor is then the leftmost vertex of its right subtree, else an ancestor), so $d(p)$ is the number of right edges of $T(p)$.

Let $W_t(n,k)$ count t-stack sortable n-permutations with k descents. Reverse and complement do not preserve t-stack sortability (213, 132), yet $W_t(n,k) = W_t(n,n-1-k)$.

Definition 14.28 (The map f)

Define f on permutations (of any finite set of integers) by $f(1) = 1$ and, with n the maximum: $f(LnR) = f(L)\,n\,f(R)$ if L, R are nonempty; $f(Ln) = n\,f(L)$; $f(nR) = f(R)\,n$.

Example 14.29

$f(123) = 3\,f(12) = 3\,2\,f(1) = 321$, so $f(4123) = f(123)\,4 = 3214$.

Example 14.30

$f(1423) = f(1)\,4\,f(23) = 1432.$

Example 14.31

$f(412395867) = f(4123)\,9\,f(5867) = 3214\,9\,5876.$

Proposition 14.32

$d(p) + d(f(p)) = n - 1$ for every n-permutation p.

Proof. Induction on n; $n = 1$ is trivial.

Step 1 ($p = LnR$, $L, R \neq \emptyset$, n in position i). $d(p) = d(L)+d(R)+1$, $d(f(p)) = d(f(L))+d(f(R))+1$, and $d(L) + d(f(L)) = i - 2$, $d(R) + d(f(R)) = n - i - 1$: total $n - 1$.

Step 2 ($p = Ln$). $d(L) + d(f(L)) + 1 = (n - 2) + 1$.

Step 3 ($p = nR$). $1 + d(R) + d(f(R)) = 1 + (n - 2)$. ■

Lemma 14.33

$s(p) = s(f(p))$ for every permutation p.

Proof. Induction, Lemma 14.22, and $s(L) = s(f(L))$, $s(R) = s(f(R))$:

(1) $p = LnR$: $s(p) = s(L)s(R)n = s(f(L))s(f(R))n = s(f(L)nf(R)) = s(f(p))$.

(2) $p = nR$: $s(p) = s(R)n = s(f(R))n = s(f(R)n) = s(f(p))$.

(3) $p = Ln$: $s(p) = s(L)n = s(f(L))n = s(nf(L)) = s(f(p))$. ■

Corollary 14.34

p is t-stack sortable iff $f(p)$ is.

Proof. Both mean $s(p) = s(f(p))$ is $(t - 1)$-stack sortable. ■

Theorem 14.35

$W_t(n, k) = W_t(n, n - 1 - k)$ for all n, k, t.

Proof. f is a bijection (on $T(p)$ it flips every only child, an involution). By Corollary 14.34 and Proposition 14.32 it maps t-stack sortable permutations with k descents onto those with $n - 1 - k$. ■

Tree view. f flips every only child and keeps two-child vertices, swapping left and right edge counts (Proposition 14.32); postorder ignores the side of an only child (Lemma 14.33), at every subtree.

Example 14.36

$p = 356124$ and $f(p) = 536421$ both give 351246 in postorder, also on each subtree.

Chapter 15. The Probabilistic Method

15.1 The notion of probability

Motivation. Finite probability is counting over counting; the union bound drives the next section.

Four coin tosses: 5 of 16 outcomes have at least three heads, probability 5/16.

Definition 15.1 (Probability)

Let Ω be a finite set of equally likely outcomes (the **sample space**) and $A \subseteq \Omega$ an **event**. The **probability** of A is $P(A) = \frac{|A|}{|\Omega|}$.

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (sieve); the **complement** $\bar{A} = \Omega - A$ has $P(\bar{A}) = 1 - P(A)$.

Proposition 15.2 (Union bound)

For events $A_1, \dots, A_n$, $P(A_1 \cup \cdots \cup A_n) \le P(A_1) + \cdots + P(A_n)$.

Proof. $|A_1 \cup \cdots \cup A_n| \le |A_1| + \cdots + |A_n|$, since the right side counts each element at least once. ■

Example 15.3

Six numbers are drawn from [36]. A ticket of six numbers contains at least one winning number with probability

$$1 - \binom{30}{6} \Big/ \binom{36}{6} \approx 1 - 0.3048 = 0.6952,$$

not $\frac{6}{36}$.

Example 15.4 (Birthday paradox)

Among 23 people (nobody born on February 29), two share a birthday with probability more than $\frac{1}{2}$, since

$$\frac{365 \cdot 364 \cdots 343}{365^{23}} < \frac{1}{2}.$$

15.2 Non-constructive proofs

Motivation. If a random object is bad with probability < 1, a good one exists.

Theorem 15.5 (Erdős)

$R(k,k) > 2^{k/2}$ for $k \ge 3$.

Proof. Let $n \le 2^{k/2}$ and colour each edge of K_n by an independent fair coin. A fixed K_k is monochromatic with probability $2^{1-\binom{k}{2}}$; let A_S be this event for S. By Proposition 15.2,

$$P\Big(\bigcup_S A_S\Big) \le \sum_S P(A_S) = \binom{n}{k} 2^{1-\binom{k}{2}}. \tag{15.1}$$

With $\binom{n}{k} < \frac{n^k}{k!}$ and $n^k \le 2^{k^2/2}$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{2^{k^2/2}}{k!} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2 \cdot 2^{k/2}}{k!} < 1$$

for $k \ge 3$ (at $k = 3$ it is $\frac{2 \cdot 2^{3/2}}{6}$; $k \mapsto k+1$ multiplies by $\frac{\sqrt{2}}{k+1} < 1$). So some colouring has no monochromatic K_k. ■

With Corollary 13.8: $(\sqrt{2})^k < R(k,k) \le 4^{k-1}$, essentially the best known bounds apart from $(4-\epsilon)^k$.

Theorem 15.6

Let $n, m \ge 2$ with $m \ge 2 \log_2 n$. Then the edges of $K_{n,n}$ can be 2-coloured with no monochromatic $K_{m,m}$.

Proof. Colour randomly: each of the $\binom{n}{m}^2$ copies of $K_{m,m}$ is monochromatic with probability $2 \cdot 2^{-m^2}$. It suffices that $2\binom{n}{m}^2 < 2^{m^2}$: $\binom{n}{m} \le \frac{n^m}{m!}$, $m! \ge 2$ give $2\binom{n}{m}^2 < n^{2m}$, and $n \le 2^{m/2}$ gives $n^{2m} \le 2^{m^2}$. ■

Equivalently, some $n \times n$ 0–1 matrix has no constant $m \times m$ submatrix, $m \approx 2 \log_2 n$; explicit constructions only reach $m = c\sqrt{n}$.

15.3 Independent events

Motivation. Independence: learning B does not change the odds of A.

15.3.1 The notion of independence and Bayes' theorem

For x uniform on $[12]$, divisibility by 2 and by 3 satisfy the product rule; by 2 and by 4 do not.

Definition 15.7 (Independence)

Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$, and **dependent** otherwise.

Definition 15.8 (Conditional probability)

For $P(B) > 0$, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ is the probability of A given B.

Proposition 15.9

For $P(B) > 0$, A and B are independent iff $P(A \mid B) = P(A)$.

Proof. Divide $P(A \cap B) = P(A)P(B)$ by $P(B)$, or multiply back. ■

Example 15.10

Four coin tosses, at least two heads (B). The probability of four heads is $P(A \mid B) = \frac{P(A)}{P(B)} = \frac{1/16}{11/16} = \frac{1}{11}$ (with $P(B) = \frac{1+4+6}{16}$).

Example 15.11

For a random n-permutation, $A = \{p_1 > p_2\}$, $B = \{p_2 > p_3\}$: $P(A) = P(B) = \frac{1}{2}$, $P(A \cap B) = \frac{1}{6}$, so $P(A \mid B) = \frac{1}{3} \neq P(A)$: dependent.

Example 15.12 (Simpson's paradox)

Both colleges admit domestic applicants at a higher rate, the university does not:

	Liberal Arts	Engineering	University
Domestic	10/120 (8.3%)	10/10 (100%)	20/130 (15.4%)
International	1/15 (6.7%)	90/100 (90%)	91/115 (79.1%)

87% of international but 8% of domestic applicants chose the easy Engineering.

Theorem 15.13 (Bayes' theorem, law of total probability)

If A, B are mutually exclusive with $A \cup B = \Omega$ and C is any event, then

$$P(C) = P(C \mid A)P(A) + P(C \mid B)P(B). \tag{15.2}$$

Proof. $P(C) = P(C \cap A) + P(C \cap B)$ and $P(C \cap A) = P(C \mid A)P(A)$, likewise for B. ■

$P(C)$ is a weighted average; in Example 15.12 the weights differ between the groups.

Example 15.14

Exactly one of Andrew (A) and Brandi (B) is guilty; the culprit has a blood type T shared by 5%. If Andrew has type T (X),

$$P(A \mid X) = \frac{P(A \cap X)}{P(X)} = \frac{P(X \mid A)P(A)}{P(X)}. \tag{15.3}$$

$P(X \mid A) = 1$, $P(X \mid B) = \frac{1}{20}$, $P(A) = P(B) = \frac{1}{2}$, so $P(X) = \frac{21}{40}$ by (15.2) and $P(A \mid X) = \frac{20}{21}$.

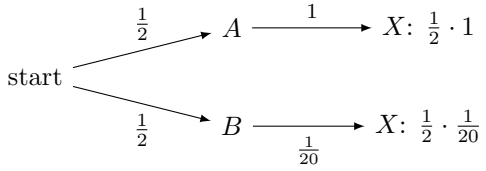


Figure. *Example 15.14 as a tree:* $P(A \mid X) = \frac{1/2}{1/2+1/40} = \frac{20}{21}$.

15.3.2 More than two events

Pairwise independence is not enough: for $x \in [10]$, $y \in [20]$, $A = \{x \text{ odd}\}$, $B = \{y \text{ odd}\}$, $C = \{x - y \text{ odd}\}$ are pairwise independent with probability $\frac{1}{2}$, but $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.

Definition 15.15 (Mutual independence)

$A_1, \dots, A_n$ are **independent** if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every nonempty $\{i_1, \dots, i_k\} \subseteq [n]$.

Theorem 15.16 (Bayes' theorem, general version)

If $A_1, \dots, A_n$ are pairwise disjoint with union Ω and C is any event, then $P(C) = \sum_{i=1}^{n} P(C \mid A_i)P(A_i)$ (terms with $P(A_i) = 0$ omitted).

Proof. $C = \bigsqcup_i (C \cap A_i)$ and $P(C \cap A_i) = P(C \mid A_i)P(A_i)$. ∎

Monty Hall. Door 1 is picked; the host opens door 2 showing a goat (B). With $A_i =$ "car behind door i",

$$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{\sum_i P(B \mid A_i)P(A_i)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{1}{3}, \tag{15.4}$$

so switching wins with probability $\frac{2}{3}$.

15.4 Expected values

Motivation. Linearity splits complicated counts into indicators; some outcome beats the average.

A **random variable** is a function $X : \Omega \to \mathbb{R}$.

Definition 15.17 (Expectation)

If X takes finitely many values S, its **expected value** is $E(X) = \sum_{i \in S} i\, P(X = i)$, where $P(X = i) = \frac{|\{u : X(u) = i\}|}{|\Omega|}$. Equivalently,

$$E(X) = \frac{1}{|\Omega|} \sum_{u \in \Omega} X(u). \tag{15.5}$$

(Write E_Ω if needed; countable ranges need convergent sums.)

Definition 15.18 (Independent variables)

X, Y are **independent** if $P(X = s, Y = t) = P(X = s)P(Y = t)$ for all s, t.

15.4.1 Linearity of expectation

Theorem 15.19 (Linearity)

(1) $E(X + Y) = E(X) + E(Y)$ for random variables on the same space.
(2) $E(cX) = cE(X)$ for $c \in \mathbb{R}$.

Proof.

Step 1 (equally likely outcomes). By (15.5): $\sum_u (X + Y)(u) = \sum_u X(u) + \sum_u Y(u)$, $\sum_u cX(u) = c \sum_u X(u)$.

Step 2 (without uniformity). With values x_i, y_j,

$$E(X+Y) = \sum_{i,j}(x_i+y_j)P(X=x_i, Y=y_j) = \sum_i x_i P(X=x_i) + \sum_j y_j P(Y=y_j);$$

for (2), $P(cX = cx_i) = P(X = x_i)$ if $c \neq 0$. ■

No independence is needed. An **indicator variable** of an event has expectation its probability.

i is a **valley** of p if $2 \le i \le n-1$ and $p_i < p_{i-1}$, $p_i < p_{i+1}$.

Theorem 15.20

For $n \ge 2$, a random n-permutation has on average $\frac{n-2}{3}$ valleys.

Proof. The indicator Y_i of a valley at i has $E(Y_i) = \frac{1}{3}$ (each of p_{i-1}, p_i, p_{i+1} equally likely smallest), so $E(\sum_{i=2}^{n-1} Y_i) = \frac{n-2}{3}$. ■

Theorem 15.21

A random n-permutation has on average one fixed point.

Proof. The indicator X_i of $p_i = i$ has $E(X_i) = \frac{1}{n}$, so

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot \frac{1}{n} = 1. \tag{15.6}$$

■

15.4.2 Existence proofs using expectation

Theorem 15.22

If X takes finitely many values with maximum j, then $j \ge E(X)$. Hence some outcome u has $X(u) \ge E(X)$.

Proof. $E(X) = \sum_{i \in S} i\, P(X=i) \le j \sum_{i \in S} P(X=i) = j$. ■

Theorem 15.23

A simple graph G on $[n]$ ($n \ge 2$) with $m \ge 1$ edges has a bipartite subgraph with more than $m/2$ edges.

Proof. Let Ω be the $2^{n-1}-1$ splits $\{A, B\}$ of $[n]$ into nonempty parts and X the number of crossing edges. Edge uv crosses in 2^{n-2} splits (place the other $n-2$ vertices freely), so

$$E(X_i) = P(X_i = 1) = \frac{2^{n-2}}{2^{n-1}-1} > \frac{1}{2}.$$

Then $E(X) > \frac{m}{2}$; apply Theorem 15.22. ■

Example 15.24 (Betweenness)

Given triples $L_i = (a_i, b_i, c_i)$ of distinct elements of $[n]$, $i \le k$, say that p satisfies L_i if b_i lies between a_i and c_i in p. Some n-permutation satisfies at least $k/3$ of the triples.
Solution. For random p, L_i holds with probability $\frac{1}{3}$; the expected number satisfied is $\frac{k}{3}$ (Theorem 15.22).

15.4.3 Conditional expectation

$E(X \mid A) = \sum_i i\, P(X = i \mid A)$ for $P(A) > 0$.

Theorem 15.25

If $A_1, \dots, A_n$ are pairwise disjoint with union Ω (and positive probabilities), then $E(X) = \sum_{i=1}^{n} E(X \mid A_i) P(A_i)$.

Proof. Theorem 15.16 with $C = \{X = j\}$, times j, summed over j:

$$E(X) = \sum_j j \sum_i P(X = j \mid A_i) P(A_i) = \sum_i P(A_i) \sum_j j\, P(X = j \mid A_i) = \sum_i E(X \mid A_i) P(A_i).$$

■

Example 15.26

Three die throws, the first ≥ 4. It is even (A_1) with probability $\frac{2}{3}$, giving on average $1 + 2 \cdot \frac{1}{2} = 2$ even throws, and odd (A_2) with $\frac{1}{3}$, giving 1. Expected even throws: $\frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 1 = \frac{5}{3}$.

Example 15.27

A team wins each game independently with probability $\frac{3}{4}$. Given at least three wins in the first four games (B), what is the expected number of wins in 12 games?
Solution. A_1, A_2: exactly 3 / all 4 first games won. $P(A_1) = 4 \cdot \frac{1}{4}(\frac{3}{4})^3 = \frac{27}{64}$, $P(A_2) = (\frac{3}{4})^4 = \frac{81}{256}$, so $P_B(A_1) = \frac{4}{7}$, $P_B(A_2) = \frac{3}{7}$; the last eight add 6 on average. By Theorem 15.25 in B,

$$E_B(X) = 9 \cdot \frac{4}{7} + 10 \cdot \frac{3}{7} = \frac{66}{7},$$

more than the unconditional 9.

15.5 Selected exercises

Motivation. Indicators for an existence proof and for a second moment.

Exercise 15.17 (Tournaments with many Hamiltonian paths)

Prove that for every $n \ge 1$ there is a tournament on n vertices with at least $\frac{n!}{2^{n-1}}$ Hamiltonian paths. What can be said about Hamiltonian cycles?

Solution.

Step 1 (random tournament). Orient each edge of K_n independently with probability $\frac{1}{2}$ each way.

Step 2 (indicators). For a permutation σ, the indicator X_σ of the path $\sigma_1 \to \cdots \to \sigma_n$ fixes $n-1$ edge directions: $E(X_\sigma) = 2^{-(n-1)}$.

Step 3 (expectation). $E(\sum_\sigma X_\sigma) = \frac{n!}{2^{n-1}}$ (Theorem 15.19); apply Theorem 15.22.

Cycles ($n \geq 3$): $(n-1)!$ cyclic orders (Example 6.10), each fixing n directions, give a tournament with $\geq \frac{(n-1)!}{2^n}$ Hamiltonian cycles. ■

Exercise 15.18 (The variance of the number of fixed points)

For a random variable Y let $\mathrm{Var}(Y) = E((Y - E(Y))^2)$.
(a) Prove that $\mathrm{Var}(Y) = E(Y^2) - E(Y)^2$.
(b) Let $X(p)$ be the number of fixed points of a random n-permutation p, $n \geq 2$. Prove that $\mathrm{Var}(X) = 1$.

Solution.

(a) With $\mu = E(Y)$, by Theorem 15.19,

$$E((Y-\mu)^2) = E(Y^2) - 2\mu E(Y) + \mu^2 = E(Y^2) - \mu^2.$$

(b)

Step 1 (indicators). $X = \sum_i X_i$, X_i the indicator of $p(i) = i$, $E(X_i) = \frac{1}{n}$, $E(X) = 1$.

Step 2 (second moment). As $X_i^2 = X_i$,

$$X^2 = \sum_i X_i + \sum_{i \neq j} X_i X_j,$$

and $E(X_i X_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$ for $i \neq j$, so

$$E(X^2) = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 2.$$

Step 3 (conclusion). $\mathrm{Var}(X) = 2 - 1^2 = 1$ (like Poisson(1); for $n = 1$ it is 0). ■